

Fisher–Rao-Guided Generative Steganography: Geodesic Embedding and Detection Analysis

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ABSTRACT

Generative and coverless steganography can be viewed as a statistical signal-generation problem: a message is embedded by steering a generator from a cover distribution toward a message-bearing distribution while preserving low detectability. Existing secure and coverless schemes primarily control the endpoint divergence between cover and stego distributions, but they do not model the detectability accumulated along the embedding trajectory. We introduce a Fisher–Rao-guided embedding framework in which the generator parameter space is treated as a statistical signal manifold and embedding paths are evaluated by their sequential Kullback–Leibler cost. In the small-step regime, we prove that this sequential detectability is governed by Fisher path energy and that Fisher–Rao geodesics are locally optimal embedding trajectories. We then derive Natural-Gradient Steganography (NG-Stego), analyze its convergence under geodesic smoothness and strong-convexity assumptions, and relate the optimal Neyman–Pearson steganalyzer in exponential families to a dual-affine information projection. Scalar and sectional curvature are used as intrinsic diagnostics of embedding path instability and local geometric sensitivity. Numerical experiments on Gaussian and Gaussian-mixture signal models illustrate geodesic sequential-KL reduction, natural-gradient speedups over Euclidean and adaptive baselines, and curvature-driven robustness effects. The result is not an operational image steganography pipeline, but a signal-processing methodology for designing and analyzing generative steganographic embedding paths.

1. Introduction

1.1. Background and Motivation

The information-theoretic security of a steganographic system was placed on firm mathematical foundations by Cachin [1], who proposed measuring the *steganographic detectability* of an embedding scheme $\mathcal{E}$ by the Kullback–Leibler (KL) divergence between the cover distribution P_C and the stego distribution P_S :

$$\varepsilon(\mathcal{E}) = D_{\text{KL}}(P_S \parallel P_C) = \int P_S(x) \log \frac{P_S(x)}{P_C(x)} dx. \quad (1)$$

A scheme is *perfectly secure* if $\varepsilon(\mathcal{E}) = 0$ and ε -*secure* if $\varepsilon(\mathcal{E}) \leq \varepsilon$. This elegant criterion has driven two decades of theoretical and empirical research in steganography.

Nevertheless, the steganographic community has employed the KL divergence solely as a scalar cost function, without exploiting the rich geometric structure in which it naturally lives. Since the seminal work of Rao [2] and Fisher [3], and the systematic development by Amari [4, 5], it is known that the space of parametric probability distributions is *not* a flat Euclidean space: it carries the structure of a *Riemannian manifold*, the *statistical manifold*, whose intrinsic geometry is encoded by the *Fisher information matrix*.

Crucially, the KL divergence is *not* a distance on this manifold, as it fails symmetry and the triangle inequality, but its Hessian at the diagonal coincides with the Fisher metric, making information geometry the natural language for studying trajectories in distribution space. When a coverless generator transforms a source distribution into a message-carrying distribution, it traces a *path* in the statistical manifold; the

accumulated detectability along this path depends on both the endpoints *and the trajectory itself*. Current methods ignore the trajectory entirely. This limitation is especially relevant for modern generative and coverless steganography, where embedding is naturally realized as controlled transport through a latent or parameter space rather than as isolated pixel-level modification. From a signal-processing viewpoint, this changes the object of analysis: the embedding operation is no longer only a final distributional mismatch, but a controlled trajectory in a statistical signal model. The Fisher–Rao metric is used here not as an end in itself, but as the intrinsic sensitivity metric that connects infinitesimal parameter perturbations to local changes in the generated signal distribution.

1.2. Coverless Steganography and the Role of Generative Models

Coverless steganography [6] encodes information by selecting or generating cover objects whose statistical properties natively encode the message, rather than by modifying an existing cover. The embedding process is then parameterized by the generator’s parameter space Θ : encoding a message m amounts to navigating from a “neutral” parameter configuration θ_0 to a “message-encoding” configuration θ_1 .

State-of-the-art coverless methods include SparSamp [7], iMEC [8], and Discop [9]. These methods focus on secure distribution matching, coupling, sampling, or decoding objectives at the level of the generated object or endpoint distribution. They do not formulate embedding as a Fisher–Rao path-cost problem and do not use (i) the intrinsic geometry of the generator parameter space, (ii) preferred (geodesic) embedding trajectories, or (iii) the curvature

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of the distribution manifold as an indicator of intrinsic instability.

1.3. Fisher–Rao Tools for Steganographic Signal Embedding

In [5], information geometry, as systematized by Amari and Nagaoka and extended by Ay et al. [10], studies the Riemannian geometry of families of probability distributions equipped with the *Fisher–Rao metric*

$$g_{ij}(\theta) = F_{ij}(\theta) = \mathbb{E}_{P_\theta} \left[\frac{\partial \log p_\theta(X)}{\partial \theta^i} \frac{\partial \log p_\theta(X)}{\partial \theta^j} \right], \quad (2)$$

which endows the parameter space Θ with a canonical Riemannian structure. Its central objects are:

1. the *Fisher–Rao metric* $g_{ij}(\theta)$, the unique (up to scale) Riemannian metric on the statistical manifold that is invariant under sufficient statistics (Chentsov’s theorem [11]);
2. the one-parameter family of *Amari–Chentsov α -connections* $\nabla^{(\alpha)}$, which interpolate between the mixture ($\alpha = -1$) and exponential ($\alpha = 1$) connections;
3. the *natural gradient* [12], the Riemannian gradient of any functional on the statistical manifold, which can improve the conditioning of gradient descent by preconditioning updates with the inverse Fisher metric; under the assumptions of Theorem 5, this yields a convergence-rate comparison controlled by the Fisher condition number.

To the best of our knowledge, the resulting path-cost formulation has not been developed for generative steganographic embedding. The present paper establishes this connection rigorously.

1.4. Core Scientific Contributions

This paper develops a steganographic signal-processing methodology for generative embedding. Fisher–Rao geometry is used as the mathematical instrument for measuring distributional sensitivity, but the scientific target is the design and analysis of low-detectability embedding trajectories in statistical signal generators. The main contributions are as follows.

1. **Detection-aware generative embedding model.** We model coverless embedding as controlled motion of a generative signal distribution from a cover state to a message-bearing state.
2. **Sequential detectability path cost.** We introduce the accumulated sequential-KL cost of a multi-step embedding path and prove that, in the small-step regime, its leading term is the Fisher path energy. Fisher–Rao geodesics therefore define locally optimal embedding trajectories for this detectability criterion.

3. **Natural-gradient embedding optimization.** We derive NG-Stego as a Fisher-preconditioned embedding optimizer and analyze its convergence under standard geodesic strong-convexity and smoothness assumptions.
4. **Curvature-based robustness diagnostics.** We study scalar curvature and minimum sectional curvature as coordinate-invariant descriptors of local path instability, metric anisotropy, and sensitivity of neighboring embedding trajectories.
5. **Detection-theoretic interpretation.** Under exponential-family assumptions, we connect Neyman–Pearson testing, information projections, and local Stein exponents to the Fisher–Rao distance between cover and stego signal distributions.
6. **Reproducible numerical validation.** We illustrate and support the theoretical predictions on Gaussian and Gaussian-mixture statistical signal models, where Fisher metrics, geodesics, curvature, and natural-gradient behavior can be inspected directly.

1.5. Signal-Processing Contribution Beyond the Mathematical Tools

Fisher–Rao geometry, natural-gradient methods, and α -connections are well-established mathematical tools. Our contribution is not that these tools exist, but that they yield a detection-aware embedding methodology for generative steganography when applied to the signal-processing problem of steering a generator while controlling statistical distinguishability. We address directly the question: *what does the framework produce for steganographic embedding and detection beyond applying established mathematical machinery?*

A new problem formulation. Every prior steganographic method — from Cachin’s ϵ -security criterion [1] to state-of-the-art coverless schemes iMEC [8] and Discop [9] — treats embedding as a point-to-point problem: it minimizes the endpoint divergence $D_{\text{KL}}(P_{\theta_1} \| P_{\theta_0})$ between the final stego distribution and the cover distribution, while ignoring how the generator parameter evolves from θ_0 to θ_1 . This paper asks: *does the trajectory through parameter space matter for the accumulated detectability of a multi-step embedding process?* We are not aware of a prior formulation of this trajectory-level question in either the steganography or the information-geometry literature. The answer, formalized by the sequential-KL path-cost functional, is yes — and the optimal path is not the Euclidean straight line.

Non-trivial results from the path formulation. The finding that Fisher–Rao geodesics minimize the accumulated sequential KL distortion (Theorem 3(iv)) does *not* follow from the standard statement that geodesics minimize Riemannian length, which is a tautology. It requires proving that the sum $\sum_k D_{\text{KL}}(P_{\gamma(t_{k+1})} \| P_{\gamma(t_k)})$, in the limit of fine discretisation, is controlled to second order by the Fisher

Table 1

Positioning of the proposed steganographic signal-processing contributions relative to prior literature. ✓ = provided; ◦ = partially or implicitly; × = not provided. Row entries in **bold** are exclusive to this paper.

Capability	Cachin / Fridrich– Ker ^a	iMEC / Discop / SparSamp ^b	Amari IG ^c	Nat. grad. ^d	This pa- per
KL endpoint security criterion	✓	✓	×	×	✓
Coverless / generative embedding	×	✓	×	×	✓
Fisher–Rao metric on param. space	×	×	✓	✓	✓
Natural-gradient optimization	×	×	◦	✓	✓
Path-cost formulation of detectability	×	×	×	×	✓
Sequential-KL geodesic optimality (Thm. 3)	×	×	×	×	✓
Detection-duality theorem (± 1 structure, Thm. 8)	×	×	×	×	✓
Curvature trade-off in the steganographic adversarial game	×	×	×	×	✓
Convergence guarantees for stego optimizer	×	×	×	✓	✓
Numerical illustrations on statistical manifolds	×	◦	×	◦	✓

^a [1, 13]. ^b [7–9]. ^c [4, 5, 10]. ^d [12, 14].

path-energy $\int \dot{\gamma}^\top F(\gamma) \dot{\gamma} dt$ — a specific asymptotic result that links the steganographic cost to the geometric structure of the manifold. We are not aware of this theorem in the natural-gradient or information-geometry literature, where the path-cost problem for KL-divergence accumulation is not usually the object of study.

A structural duality specific to steganography. Theorem 8 shows that the optimal Neyman–Pearson steganalyzer in the exponential-family setting admits a *dual-affine* interpretation: its rejection region is affine in expectation coordinates (the m -flat half-space) and its statistic corresponds to projecting the empirical distribution along the e -geodesic direction connecting cover and stego parameters. The embedder’s optimal path lies in the *Levi-Civita* (Fisher–Rao) geometry; the detector’s optimal strategy lies in the *e-geometry*. This ± 1 structural duality, arising from the Amari–Chentsov connection pair, is specific to the adversarial steganographer–steganalyzer setting; we are not aware of an equivalent formulation in prior work on natural-gradient optimization, exponential families, or steganographic security.

The two faces of curvature. In the regime of negative sectional curvature — which includes the Gaussian family ($\kappa = -1/2$) — curvature plays opposite roles for the two adversaries simultaneously: it expands the Fisher-ball volume available to the steganalyzer (a detection advantage via Jacobi-field growth) while amplifying geodesic transport instability for the embedder (an embedding cost). This trade-off, intrinsic to the geometry of the adversarial game, appears to be absent from standard steganography and Riemannian-geometry treatments of these tools.

Summary positioning. Table 1 maps each contribution against the prior literature, making explicit what each body of prior work does and does not provide.

1.6. Organization and Scope Statement

Section 2 reviews information-theoretic steganography, coverless generative embedding, information geometry, and natural-gradient optimization. Section 3 recalls the Riemannian and statistical-geometric notions used throughout the paper. Section 4 defines the steganographic statistical manifold and derives Fisher metrics for representative generator families. Section 5 develops the geodesic embedding formulation, its local sequential-KL optimality, and the dual affine structure (α -connections, Pythagorean identity) connecting embedding transport, Riemannian optimization, and idealized detection. Section 6 introduces NG-Stego and its constrained variant. Section 7 studies scalar and sectional curvature as local geometric instability indicators. Section 8 presents synthetic numerical illustrations on Gaussian and Gaussian-mixture manifolds. Section 9 provides the detection-theoretic interpretation and scope limitations. The final section concludes and outlines future work.

Scope. The framework is methodological and theoretical. It is designed to clarify how generative steganographic signals can be embedded, optimized, and analyzed in distribution space, not to provide a complete operational image-steganography pipeline. All geodesic and detection statements are local, asymptotic, or model-dependent unless explicitly stated otherwise. Extensions to high-dimensional deep generators, full payload-rate analysis, and modern steganalysis benchmarks are left for future work.

2. Related Work

2.1. Information-Theoretic Steganography and Steganalysis

The mathematical foundations of steganographic security were laid by Cachin [1], who formalised the ϵ -security criterion via the KL divergence between cover and stego distributions. This information-theoretic framework was later refined by Fridrich and Ker [13], who derived asymptotic bounds on detectability under empirical cover models.

On the detection side, Pevný, Filler and Bas [15] introduced high-dimensional image models for steganalysis, demonstrating that the KL divergence remains the dominant statistical signal even when the adversary employs rich feature representations. These works treat the KL divergence as a *scalar* cost evaluated at the endpoints of the embedding process; they do not consider the *trajectory* in distribution space, nor the Riemannian geometry induced by the Fisher metric along that trajectory. Consequently, the accumulated detectability of a multi-step embedding path is approximated by its endpoint divergence, which can be suboptimal whenever the chosen coordinate path fails to be a Fisher–Rao geodesic; nonzero curvature is one source of such path dependence but is not, by itself, a sufficient condition.

Modern steganalysis has moved beyond hand-crafted features to deep learning detectors, including rich models [16] and residual-network classifiers [17], which approach near-optimal discrimination for empirical cover sources. From an information-theoretic standpoint, the capacity and fundamental limits of steganography and information hiding have been characterized by Moulin and O’Sullivan [18]. Practical embedding schemes often minimize additive distortion via syndrome-trellis codes [19]. These methods optimize endpoint statistics or distortion, not the generator trajectory in parameter space. Geometric approaches to signal and tensor analysis, notably the Fisher–Rao geometry of covariance matrices [20], provide complementary tools for the parametric trajectory analysis developed in the present paper.

2.2. Coverless Steganography via Generative Models

Coverless steganography eliminates the cover-modification paradigm by generating or selecting objects whose native statistics encode the message. Recent methods exploit deep generative models. Ziegler and Rush [21] proposed neural linguistic steganography via arithmetic coding over language-model distributions; Zhang et al. [22] extended the principle to image generation with explicit rate–distortion control.

State-of-the-art provably secure schemes include iMEC [8] and Discop [9], which use distribution-preserving coupling or sampling mechanisms. These methods do not optimize a Fisher–Rao trajectory-level embedding cost or use the natural gradient induced by the FIM for that purpose. From the perspective of information geometry, any prescribed coordinate path in Θ may incur higher accumulated Fisher energy than a length-minimizing Fisher–Rao geodesic when it fails to satisfy the Fisher–Rao geodesic equation.

2.3. Information Geometry and Natural Gradient Optimisation

The differential-geometric structure of statistical manifolds was systematised by Rao [2] and Fisher [3], and developed into the modern framework of information geometry by Amari [4, 5, 12] and Chentsov [11]. Ay et al. [10] and Nielsen [23] provide comprehensive modern treatments.

The natural gradient [12] replaces Euclidean updates with Riemannian descent on the Fisher metric. For deep neural networks, Martens and Grosse [14] introduced Kronecker-factored approximate curvature (K-FAC), making natural-gradient training tractable for million-parameter models. Bonnabel [24] analysed stochastic gradient descent on Riemannian manifolds with retractions, providing the convergence guarantees underlying Algorithm 2.

2.4. Geometric Methods in Statistical Learning and Transport

Riemannian geometry has been applied to signal processing on the cone of positive-definite covariance matrices [25] and to shape analysis, yet not to the steganographic embedding problem. Optimal transport (OT) theory [26, 27] has gained traction in machine learning as an alternative to KL-based divergence measures.

Important distinction. The Wasserstein framework operates on the *sample space* $\mathcal{X}$ rather than the *parameter space* Θ of the generator. For parametric families with smooth densities, the Fisher–Rao metric provides a *different* geometry than the Wasserstein metric: it is intrinsic to the statistical model, invariant under sufficient statistics (Chentsov’s theorem), and its geodesics locally minimize the leading-order accumulated sequential-KL cost along the transport path in the small-step regime (Theorem 3). The two geometries are complementary, not competing: Fisher–Rao describes the parameter-space geometry of the embedding operation, while Wasserstein describes the sample-space geometry of the generated distributions.

A research programme even closer to the present setting is outlined by Bharadwaj and Moskowitz [28], who propose to apply Fisher–Rao geometry and statistical manifolds to image analysis and steganography. That work, however, remains at the conceptual level: it does not formulate a sequential embedding cost, does not derive geodesic embedding optimality, does not connect embedding transport to Neyman–Pearson detection, and does not analyse curvature as an embedder–detector trade-off. The present paper develops precisely these formal elements.

2.5. Positioning and Gaps

While individual elements of our framework (Fisher metric, natural gradient, geodesic transport) have been studied extensively in statistics and machine learning, their *joint application* to the design and security analysis of steganographic systems appears to be new. Existing coverless methods lack (i) a geometric characterisation of optimal embedding paths, (ii) a curvature-based instability analysis, and (iii) a Riemannian optimisation algorithm tailored to the KL objective. The present paper unifies these ingredients under a single information-geometric formalism, with explicit scope limitations (Section 10.5).

Related studies of gradient systems on statistical manifolds, including the complete integrability of gradient flows on lognormal and beta families, develop Hessian-metric structures on exponential families closely related to those used in the present paper.

3. Mathematical Background

3.1. Riemannian Manifolds

We recall standard notions from Riemannian geometry; see, for example, [25, 29, 30].

Definition 1. A smooth Riemannian manifold $(\mathcal{M}, g)$ is a smooth manifold $\mathcal{M}$ together with a Riemannian metric g , i.e., a family of positive-definite inner products $g_p : T_p\mathcal{M} \times T_p\mathcal{M} \rightarrow \mathbb{R}$ on each tangent space $T_p\mathcal{M}$, varying smoothly with $p \in \mathcal{M}$.

In local coordinates $\theta = (\theta^1, \dots, \theta^d)$, the metric is represented by the symmetric positive-definite matrix $\mathbf{G}(\theta) = [g_{ij}(\theta)]$, where $g_{ij}(\theta) = g\left(\frac{\partial}{\partial\theta^i}, \frac{\partial}{\partial\theta^j}\right)$. The Riemannian length of a smooth curve $\gamma : [0, 1] \rightarrow \mathcal{M}$ is

$$\ell(\gamma) = \int_0^1 \sqrt{g_{ij}(\gamma(t)) \dot{\gamma}^i(t) \dot{\gamma}^j(t)} dt.$$

Definition 2. The Levi-Civita connection ∇ of $(\mathcal{M}, g)$ is the unique torsion-free connection compatible with g . In local coordinates, its Christoffel symbols are given by

$$\Gamma_{ij}^k(\theta) = \frac{1}{2} g^{kl}(\theta) \left(\frac{\partial g_{il}}{\partial\theta^j} + \frac{\partial g_{jl}}{\partial\theta^i} - \frac{\partial g_{ij}}{\partial\theta^l} \right). \quad (3)$$

Definition 3. A smooth curve $\gamma : [0, 1] \rightarrow \mathcal{M}$ is a geodesic if it satisfies

$$\ddot{\gamma}^k + \Gamma_{ij}^k(\gamma) \dot{\gamma}^i \dot{\gamma}^j = 0, \quad k = 1, \dots, d. \quad (4)$$

Definition 4. For linearly independent tangent vectors $u, v \in T_p\mathcal{M}$, the sectional curvature of the plane spanned by u and v is

$$K(u, v) = \frac{R_{ijkl} u^i v^j u^k v^l}{(g_{ik} g_{jl} - g_{il} g_{jk}) u^i v^j u^k v^l}. \quad (5)$$

Definition 5. The scalar curvature $R(p)$ at $p \in \mathcal{M}$ is the trace of the Ricci curvature tensor with respect to the metric:

$$R(p) = g^{ij}(p) R_{ij}(p) = g^{ij}(p) R_{ikj}^k(p). \quad (6)$$

It is a coordinate-invariant scalar field on $\mathcal{M}$.

3.2. Statistical Manifold and Fisher–Rao Metric

We use standard terminology from information geometry and statistical manifold theory; see [5, 31, 32].

Let $\mathcal{X}$ be a measurable space and let $\Theta \subset \mathbb{R}^d$ be an open parameter domain. Consider a family of probability distributions $\{P_\theta : \theta \in \Theta\}$ dominated by a σ -finite measure μ , with smooth densities $p_\theta = dP_\theta/d\mu$. The log-likelihood function is $\ell(x, \theta) = \log p_\theta(x)$.

Definition 6. A statistical model on $(\mathcal{X}, \mathcal{B}, \mu)$ is a parametric family $\mathcal{S} = \{P_\theta : \theta \in \Theta\}$ of μ -dominated probability measures whose densities $p_\theta = dP_\theta/d\mu$ are smooth in θ and for which the Fisher information matrix exists and is positive definite on Θ .

Definition 7. A statistical manifold is a triplet $(\mathcal{S}, g, \nabla)$ where g is a Riemannian metric and ∇ is a torsion-free affine connection such that ∇g is totally symmetric. In this work, g is taken to be the Fisher–Rao metric. The Levi-Civita connection $\nabla^{(0)}$ and the Amari–Chentsov α -connections $\nabla^{(\alpha)}$ are the canonical instances considered later.

The Fisher information matrix is defined as follows; see [2, 5].

Definition 8. The Fisher information matrix at $\theta \in \Theta$ is the $d \times d$ symmetric matrix

$$\begin{aligned} F_{ij}(\theta) &= \mathbb{E}_{P_\theta} \left[\frac{\partial \log p_\theta(X)}{\partial \theta^i} \frac{\partial \log p_\theta(X)}{\partial \theta^j} \right] \\ &= -\mathbb{E}_{P_\theta} \left[\frac{\partial^2 \log p_\theta(X)}{\partial \theta^i \partial \theta^j} \right]. \end{aligned} \quad (7)$$

The following uniqueness result is due to Chentsov [11].

Theorem 1. Consider the space $\mathcal{P}$ of all probability measures on a measurable space, equipped with the structure of statistical manifolds. The Fisher information matrix $F(\theta)$ is, up to a positive scalar factor, the unique Riemannian metric on $\mathcal{P}$ that is invariant under all Markov morphisms, also called congruent embeddings.

Remark 1. Chentsov’s theorem applies to the entire space of probability distributions, not to a fixed parametric family $\{P_\theta : \theta \in \Theta\}$. For such a family, the invariance group is restricted to Markov morphisms that preserve the family structure, and uniqueness is not guaranteed: other metrics may exist that are invariant under this smaller group. The Fisher metric remains the canonical choice, as it is the only metric that extends uniquely to the ambient space, but it is not the unique invariant metric within the family.

This canonical status, crucially absent from Euclidean metrics, motivates defining the geometry of $\mathcal{S}_\mathcal{G}$ via the FIM. Henceforth, we set $g_{ij}(\theta) = F_{ij}(\theta)$ and refer to it as the Fisher–Rao metric.

3.3. KL Divergence and the Fisher Metric

The following second-order expansion is standard in information geometry under the usual regularity assumptions allowing differentiation under the integral sign.

Proposition 1. For $\theta' \rightarrow \theta$, the KL divergence admits the second-order Taylor expansion

$$D_{\text{KL}}(P_{\theta'} \| P_\theta) = \frac{1}{2} (\theta' - \theta)^\top F(\theta) (\theta' - \theta) + O(\|\theta' - \theta\|^3). \quad (8)$$

3.4. Natural Gradient

The natural gradient was introduced by Amari as the Riemannian gradient induced by the Fisher information metric [12].

Definition 9. Let $\ell : \Theta \rightarrow \mathbb{R}$ be a smooth functional. The natural gradient of ℓ at θ is

$$\tilde{\nabla}_\theta \ell(\theta) = F^{-1}(\theta) \nabla_\theta \ell(\theta). \quad (9)$$

4. The Steganographic Manifold

4.1. Definition and Basic Properties

Let $\mathcal{G}$ be a coverless generator with parameter space $\Theta \subset \mathbb{R}^d$, producing objects in $\mathcal{X}$ with distribution P_θ for each $\theta \in \Theta$.

Assumption 1 (Regularity). (i) Θ is an open, connected subset of $\mathbb{R}^d$.

(ii) For each $x \in \mathcal{X}$, the map $\theta \mapsto p_\theta(x)$ is twice continuously differentiable.

(iii) For each $\theta \in \Theta$, the Fisher information matrix $F(\theta)$ is strictly positive definite.

(iv) The map $\theta \mapsto F(\theta)$ is smooth (C^∞).

Remark 2. Assumption 1(ii) requires the density $p_\theta(x)$ to be C^2 in θ . For neural-network generators with ReLU activations, the score $\nabla_\theta \log p_\theta(x)$ is not differentiable everywhere in θ (the non-linearity is piecewise linear). Two standard remedies apply:

1. Mollification: replace ReLU by a smooth approximation $\text{softplus}_\beta(x) = \beta^{-1} \log(1 + e^{\beta x})$ with $\beta \gg 1$; the resulting density satisfies Assumption 1(ii) and converges uniformly to the ReLU network as $\beta \rightarrow \infty$.
2. Almost-everywhere differentiability: ReLU networks are differentiable except on a set of measure zero in Θ ; the Fisher metric can be defined almost everywhere, and the geodesic equations hold in a weak (Filippov) sense.

All results in this paper assume the mollified (smooth) setting; the extension to non-smooth activations is left as an open problem.

Definition 10. Under Assumption 1, the steganographic manifold associated with generator $\mathcal{G}$ is the Riemannian manifold $S_{\mathcal{G}} = (\{P_\theta : \theta \in \Theta\}, g)$, where $g_{ij}(\theta) = F_{ij}(\theta)$.

Theorem 2. Let $\mathcal{G}$ be an exponential-family generator with natural parameter space Θ open and convex, and suppose the log-partition function $\psi(\theta)$ is strictly convex with $\nabla^2 \psi(\theta) > 0$ on Θ and has super-linear growth at the boundary of Θ (i.e., $\psi(\theta) \rightarrow \infty$ faster than any linear function as θ approaches $\partial\Theta$; see Brown [33], Thm. 3.8 for the precise steepness condition). Then the steganographic manifold $S_{\mathcal{G}}$ is geodesically complete: for any two points $\theta_0, \theta_1 \in \Theta$, there exists a length-minimizing geodesic connecting them.

For the univariate Gaussian family, the geodesic is more-over unique: the Poincaré half-plane isometry gives non-positive sectional curvature ($\kappa = -1/2$), and Cartan–Hadamard then implies global uniqueness. For general exponential families, uniqueness of minimizing geodesics holds locally within the injectivity radius; global uniqueness requires non-positive sectional curvature on all of Θ , which need not hold (e.g., multinomial families can have positive curvature).

Proof. For exponential families, the Fisher metric is the Hessian of the log-partition function: $g_{ij}(\theta) = \partial_i \partial_j \psi(\theta)$. Since ψ is strictly convex, the metric g is a *Hessian metric* on the open convex set Θ .

For the univariate Gaussian family $\{\mathcal{N}(\mu, \sigma^2) : \mu \in \mathbb{R}, \sigma > 0\}$, the Fisher–Rao metric is $g = \sigma^{-2} \text{diag}(1, 2)$ (see Appendix G for the explicit derivation). The coordinate change $y = \sigma \sqrt{2}$ transforms this to $ds^2 = 2(d\mu^2 + dy^2)/y^2$, which is 2 times the Poincaré metric on $\mathbb{H}^2 = \{(\mu, y) : y > 0\}$ (see Appendix G for the factor-of-2 derivation). This constant rescaling preserves geodesics and non-positive curvature ($\kappa = -1/2$). It is well known (do Carmo [25], Prop. 2.1) that $\mathbb{H}^2$ with this metric is complete and that any two points are connected by a *unique* geodesic (vertical line or semicircle orthogonal to the real axis); Cartan–Hadamard then implies uniqueness for the Gaussian manifold.

For general exponential families with open natural parameter space $\Theta \subseteq \mathbb{R}^d$, completeness follows from the super-linear growth of ψ at the boundary of Θ (this prevents geodesics from reaching the boundary in finite distance). The result is standard; see Brown [33], Theorem 3.8. By Hopf–Rinow, metric completeness implies geodesic completeness.

Uniqueness of minimizing geodesics holds within the injectivity radius for any complete Riemannian manifold. Global uniqueness for arbitrary exponential families would require non-positive sectional curvature on all of Θ (Cartan–Hadamard), which we do not assume here. $\square$

Remark 3. Geodesic completeness is not automatic for all statistical models. A sufficient condition (Brown [33], Thm. 3.8) is that ψ is strictly convex with super-linear growth at the boundary of Θ . For models outside the exponential family (e.g., Student-t, stable distributions), completeness must be verified separately. In this paper we restrict all completeness claims to families satisfying this condition, or to the Gaussian family where the Poincaré isometry gives an independent proof.

Remark 4. Theorem 2 applies to exponential families with unconstrained natural parameters. For mixture models (e.g., GMMs) where the mixing weight $\pi \in (0, 1)$ defines a bounded parameter domain, geodesic completeness holds only within the closure of the parameter space; geodesics approaching the boundary ($\pi \rightarrow 0$ or $\pi \rightarrow 1$) may require infinite length to reach. In practice, we clamp $\pi \in [0.01, 0.99]$ to avoid this boundary effect.

4.2. Fisher Metric for Exponential Families

We recall the standard definition of exponential families following Brown [33] and Amari [5].

Definition 11. A family $\{P_\theta\}$ belongs to the exponential family if its densities take the form

$$p_\theta(x) = h(x) \exp\{\langle \theta, T(x) \rangle - \psi(\theta)\}, \quad (10)$$

where $\theta \in \Theta$ are the natural parameters, $T : \mathcal{X} \rightarrow \mathbb{R}^d$ is the sufficient statistic, $h : \mathcal{X} \rightarrow \mathbb{R}_{>0}$ is the carrier measure, and $\psi(\theta)$ is the log-partition function.

Proposition 2. For an exponential family, $F_{ij}(\theta) = \partial_i \partial_j \psi(\theta) = \text{Cov}_{P_\theta}[T_i(X), T_j(X)]$.

Example 1 (Multivariate Gaussian family). For $P_\theta = \mathcal{N}(\mu, \Sigma)$ with $\theta = (\mu, \Sigma)$, the Fisher–Rao metric decomposes as

$$F_{ij}(\mu, \Sigma) = \begin{pmatrix} \Sigma^{-1} & 0 \\ 0 & \frac{1}{2} \Sigma^{-1} \otimes \Sigma^{-1} \end{pmatrix}. \quad (11)$$

For the mean-only subfamily $\mathcal{N}(\mu, \sigma^2 I)$ with fixed σ^2 , we have $F_{ij}(\mu) = \sigma^{-2} \delta_{ij}$ (constant, diagonal). For the full family with both μ and Σ as parameters, the metric is block-diagonal but the Σ -block is non-constant (it depends on Σ^{-1}).

Example 2 (DDPM Gaussian latent process). For a Denoising Diffusion Probabilistic Model [34] parameterizing $P_\theta(x) = \mathcal{N}(x; \mu_\theta, \Sigma_\theta)$, the Fisher–Rao metric at θ is

$$F_{ij}(\theta) = \frac{\partial \mu_\theta^\top}{\partial \theta^i} \Sigma_\theta^{-1} \frac{\partial \mu_\theta}{\partial \theta^j} + \frac{1}{2} \text{tr} \left(\Sigma_\theta^{-1} \frac{\partial \Sigma_\theta}{\partial \theta^i} \Sigma_\theta^{-1} \frac{\partial \Sigma_\theta}{\partial \theta^j} \right). \quad (12)$$

Remark 5 (Signal-processing interpretation). The Gaussian families above are not merely abstract examples: they arise as the distributions of short-time Fourier coefficients, wavelet subbands, linear-prediction residuals, or other transform-domain features used in audio, image, and communications signal processing. A Gaussian mixture model can represent the marginal distribution of local image patches or speech frames. In each case, the generator parameter θ encodes second-order signal statistics, and the Fisher–Rao metric measures how sensitive the generated signal distribution is to parameter perturbations. The framework therefore applies directly to any parametric signal model whose density and score are tractable.

4.3. Fisher Metric for Latent Mixture Models

For generative models with latent variables (VAE [35], GAN [36]), the marginal density is $p_\theta(x) = \int p_\theta(x|z) p(z) dz$, which generally has no closed form. Following the empirical-Fisher approximation commonly used in natural-gradient and curvature-based optimisation, particularly in the K-FAC framework of Martens and Grosse [14], we use the following estimator.

Definition 12. Given i.i.d. samples $\{x_n\}_{n=1}^N$ from P_θ , the empirical Fisher information matrix is $\hat{F}_{ij}(\theta) = \frac{1}{N} \sum_{n=1}^N \partial_i \log p_\theta(x_n) \partial_j \log p_\theta(x_n)$.

Proposition 3. Under Assumption 1, if the score $\nabla_\theta \log p_\theta(x)$ is computed exactly (no latent variables), then $\hat{F}(\theta) \rightarrow F(\theta)$ almost surely as $N \rightarrow \infty$ by the strong law of large numbers.

For models with latent variables z , the exact score requires $\nabla_\theta \log p_\theta(x) = \mathbb{E}_{p(z|x, \theta)}[\nabla_\theta \log p_\theta(x, z)]$. With a single Monte-Carlo sample $z \sim p(z|x)$ per data point, the empirical estimator $\hat{F}$ is a stochastic approximation rather than a consistent estimator:

$$\mathbb{E}_{z|x}[\hat{F}_{ij}] = F_{ij}(\theta) + \text{Bias}_{ij}(x, \theta),$$

where $\text{Bias}_{ij}(x, \theta) = \text{Cov}_{p(z|x)}(\partial_i \log p_\theta(x, z), \partial_j \log p_\theta(x, z)) \neq 0$ in general. The bias vanishes only when multiple samples are used or when the posterior $p(z|x)$ is deterministic. In practice, $\hat{F}$ is treated as a computationally efficient approximation; see Martens [14] for variance-reduction techniques.

Scope note. Algorithm 2 (NG-Stego) uses the empirical Fisher computed from samples $x \sim P_{\theta_i}$ with exact score $\nabla_\theta \log p_{\theta_i}(x)$. This is tractable when $p_\theta(x)$ is available in closed form (e.g., Gaussian, GMM families, as in all experiments of Section 8). For most deep generative models (GANs, VAEs with intractable marginals), the exact score is not directly available, and the algorithm as written does not directly apply; structured approximations (K-FAC, diagonal) would be required. The present paper restricts theoretical and empirical results to models where the density is tractable.

In high-dimensional settings ($d \gg 1$), the full FIM is impractical. We consider structured approximations:

$$\hat{F}_{ij}^{\text{diag}}(\theta) = \delta_{ij} \hat{F}_{ii}(\theta), \quad (13)$$

$$\hat{F}^{(r)}(\theta) = \sum_{k=1}^r \lambda_k u_k u_k^\top, \quad (14)$$

where (14) is the rank- r truncation of the eigendecomposition.

5. Geodesic Embedding Paths

5.1. Formulation of the Optimal Embedding Problem

An embedding instance is a pair $(\theta_0, \theta_1) \in \Theta^2$, where θ_0 represents the “neutral” parameter configuration and θ_1 encodes the target message. The path energy of a smooth curve $\gamma : [0, 1] \rightarrow \Theta$ is

$$E[\gamma] = \int_0^1 \dot{\gamma}(t)^\top F(\gamma(t)) \dot{\gamma}(t) dt = \int_0^1 \|\dot{\gamma}(t)\|_g^2 dt. \quad (15)$$

5.2. Main Local Optimality Theorem

The following result states the variational role of Fisher–Rao geodesics in the embedding problem. The statement is deliberately local: on a complete Riemannian manifold, Hopf–Rinow guarantees the existence of minimizing geodesics between any two points, but not their global uniqueness. Uniqueness is guaranteed only inside a normal neighbourhood, or globally under stronger assumptions such as simple connectedness and non-positive sectional curvature.

Remark 6 (Non-triviality of part (iv)). The content of part (iv) below is not a tautology of the form “geodesics minimize length, and shorter paths are less detectable.” The steganographic objective — minimizing the accumulated sum $\sum_k D_{\text{KL}}(P_{\gamma(t_{k+1})} \| P_{\gamma(t_k)})$ over a discretised path — is not the Riemannian length functional; these two quantities coincide only in the limit of infinitesimally fine discretisation and under a specific second-order KL expansion. The claim is that this expansion, applied to the steganographic path

cost, produces the Fisher path-energy functional as the leading-order term, so that minimising over paths yields the geodesic equation as the Euler–Lagrange condition. This chain of reasoning is new: neither the information-geometry literature (which studies geodesics in the abstract) nor the steganography literature (which studies endpoint divergence) has previously connected these two quantities through the path-cost problem.

Theorem 3. Let (S_G, g) satisfy Assumption 1, and let $\theta_0, \theta_1 \in \Theta$.

- (i) Variational characterization. Critical points of the path-energy functional

$$E[\gamma] = \int_0^1 \dot{\gamma}(t)^\top F(\gamma(t)) \dot{\gamma}(t) dt \quad (16)$$

among smooth curves with fixed endpoints $\gamma(0) = \theta_0$ and $\gamma(1) = \theta_1$ are precisely constant-speed geodesics satisfying the geodesic equation.

- (ii) Local energy and length minimality. If θ_1 lies in a normal neighbourhood of θ_0 , then there exists a unique geodesic segment γ^* from θ_0 to θ_1 . This geodesic locally minimizes both the Riemannian length

$$\ell_g[\gamma] = \int_0^1 \sqrt{\dot{\gamma}(t)^\top F(\gamma(t)) \dot{\gamma}(t)} dt \quad (17)$$

and, among constant-speed curves with the same endpoints, the energy $E[\gamma]$.

- (iii) Global minimizers. If (S_G, g) is geodesically complete, then at least one length-minimizing geodesic exists between θ_0 and θ_1 . Any such length-minimizing geodesic minimizes $\ell_g[\gamma]$ globally among piecewise-smooth paths joining the two endpoints. If it is parametrized with constant speed on the fixed interval $[0, 1]$, it also minimizes the energy among all piecewise-smooth curves with the same endpoints. Global uniqueness is not asserted unless additional curvature and topology assumptions are imposed.

- (iv) Sequential detectability. Let $\gamma : [0, 1] \rightarrow \Theta$ be a smooth path, and consider the N -step embedding process obtained by sampling the path at times $t_k = k/N$. Then, under the local KL expansion of Proposition 1,

$$\sum_{k=0}^{N-1} D_{\text{KL}}(P_{\gamma(t_{k+1})} \| P_{\gamma(t_k)}) = \frac{E[\gamma]}{2N} + R_N(\gamma), \quad (18)$$

where $R_N(\gamma) = O(N^{-2})$ whenever the third-order derivatives of the local KL divergence are uniformly bounded along γ .

Proof. Parts (i)–(iii) are standard consequences of Riemannian variational theory. The first variation of the energy functional (16), under fixed-endpoint variations, yields the

geodesic equation. Inside a normal neighbourhood, the exponential map at θ_0 is a diffeomorphism onto its image, hence the geodesic segment connecting θ_0 to θ_1 is unique and locally minimizing. If the manifold is complete, Hopf–Rinow guarantees the existence of a globally length-minimizing geodesic, but not its uniqueness.

For part (iv), apply the second-order expansion of the KL divergence to each short segment:

$$D_{\text{KL}}(P_{\gamma(t+h)} \| P_{\gamma(t)}) = \frac{h^2}{2} \dot{\gamma}(t)^\top F(\gamma(t)) \dot{\gamma}(t) + O(h^3),$$

with $h = 1/N$. Summing over $k = 0, \dots, N-1$ gives (18). Uniform boundedness of the third-order local KL derivatives along γ gives $R_N(\gamma) = O(N^{-2})$. $\square$

Corollary 1. In a normal neighbourhood of θ_0 , the Fisher–Rao geodesic γ^* from θ_0 to θ_1 is the unique locally optimal embedding path: it minimizes the Riemannian length and, under constant-speed parametrization, the path energy. Consequently, in the small-step regime, it also minimizes the leading-order term of the total sequential detectability in (18).

Globally, the same conclusion holds for any length-minimizing geodesic, but uniqueness is not guaranteed in general. A non-geodesic coordinate path, such as a Euclidean straight line in an arbitrary chart, may incur larger Fisher energy whenever it fails to satisfy the Fisher–Rao geodesic equation. Nonzero sectional curvature alone is not sufficient to imply this failure: some Euclidean straight lines may still be geodesics in special coordinate systems or special regions of the manifold.

Steganographic consequence. For a generative embedder, the parameter trajectory is a relevant design variable whenever embedding is realized as a sequence of observable or physically meaningful operations—for example iterative sampling, MCMC transitions, or diffusion steps—or when the generator is sensitive to the order in which parameters are updated (e.g. numerical instability along non-geodesic paths). In these settings the sequential-KL cost is a surrogate for the cumulative statistical footprint of the embedding process. It should not be confused with the endpoint security criterion $D_{\text{KL}}(P_{\theta_1} \| P_{\theta_0})$: under the standard steganographic threat model in which the adversary observes only the final generated object, the endpoint KL remains the fundamental detectability measure. The Fisher–Rao geodesic minimizes the sequential surrogate locally; whether this also reduces final-object detectability depends on the specific generator and threat model.

5.3. Explicit Geodesics for Gaussian Subfamilies

Proposition 4. For the univariate Gaussian family $\{P_\theta = \mathcal{N}(\mu, \sigma^2) : \mu \in \mathbb{R}, \sigma > 0\}$, the Fisher metric is $g(\theta) = \sigma^{-2} \text{diag}(1, 2)$. The manifold is isometric (up to scale factor $\sqrt{2}$) to the Poincaré upper half-plane $\mathbb{H}^2$. Geodesics are either vertical lines ($\mu = \text{const}$) or semicircles centered on the $\sigma = 0$ axis.

Proof. The coordinate change $y = \sigma\sqrt{2}$ transforms the metric to $ds^2 = 2(d\mu^2 + dy^2)/y^2$, which is 2 times the standard Poincaré metric on $\mathbb{H}^2 = \{(\mu, y) : y > 0\}$ (see Appendix G). This map is a smooth diffeomorphism with smooth inverse $\sigma = y/\sqrt{2}$; the constant rescaling by 2 preserves geodesics (it scales lengths by $\sqrt{2}$ uniformly). Geodesics of $\mathbb{H}^2$ are well known: vertical lines and semicircles orthogonal to the boundary $y = 0$ (do Carmo [25]). $\square$

5.4. Amari–Chentsov α -Connections and Dual Affine Structure

The statistical manifold S_G carries not only the Levi-Civita connection $\nabla^{(0)}$ of the Fisher–Rao metric, but also a one-parameter family of torsion-free affine connections that encode the geometry of exponential and mixture families. We use the standard Amari–Chentsov formalism for α -connections and dual connections in information geometry; see Amari and Nagaoka [5].

Definition 13. For $\alpha \in \mathbb{R}$, the α -connection $\nabla^{(\alpha)}$ on S_G is defined in local coordinates by the Christoffel symbols

$$\Gamma_{ij,k}^{(\alpha)}(\theta) = \mathbb{E}_{P_\theta} \left[\left(\partial_i \partial_j \log p_\theta + \frac{1-\alpha}{2} \partial_i \log p_\theta \partial_j \log p_\theta \right) \partial_k \log p_\theta \right] \quad (19)$$

The cases $\alpha = 0$, $\alpha = 1$ (exponential, e -connection), and $\alpha = -1$ (mixture, m -connection) are the most important: $\nabla^{(0)}$ is the Levi-Civita connection of g ; $\nabla^{(1)}$ and $\nabla^{(-1)}$ are flat on exponential families.

Definition 14. Two connections ∇ and ∇^* on (S_G, g) are dual if for all vector fields X, Y, Z :

$$X g(Y, Z) = g(\nabla_X Y, Z) + g(Y, \nabla_X^* Z).$$

The α - and $(-\alpha)$ -connections are dual: $(\nabla^{(\alpha)})^* = \nabla^{(-\alpha)}$, and in particular $\nabla^{(1)}$ (e -connection) and $\nabla^{(-1)}$ (m -connection) form a dual pair with respect to the Fisher–Rao metric.

For exponential families, the three canonical geodesic types have explicit coordinate descriptions:

- e -geodesics ($\nabla^{(1)}$): straight lines in natural coordinates, $\gamma_e(t) = (1-t)\theta_0 + t\theta_1$.
- m -geodesics ($\nabla^{(-1)}$): straight lines in expectation coordinates $\eta = \nabla\psi(\theta)$, $\gamma_m(t) = \nabla\psi^{-1}((1-t)\eta_0 + t\eta_1)$.
- Riemannian geodesics ($\nabla^{(0)}$): curved in general in both coordinate systems; computed numerically via Algorithm 1.

The following generalized Pythagorean identity is a classical result of dually flat information geometry; see Amari and Nagaoka [5].

Theorem 4. Let $P, Q, R \in S_G$ be such that the e -geodesic from Q to R and the m -geodesic from P to Q meet orthogonally at Q with respect to the Fisher–Rao metric. Then:

$$D_{\text{KL}}(P \parallel R) = D_{\text{KL}}(P \parallel Q) + D_{\text{KL}}(Q \parallel R). \quad (20)$$

Algorithm 1: Steganographic Geodesic Solver (RK4-Geo)

Input: Initial position θ_0 , initial velocity $v_0 \in T_{\theta_0} S_G$, step size Δt , number of steps N_t .

Output: Trajectory $(\theta_0, \theta_1, \dots, \theta_{N_t})$.

```

1 for  $k = 0, \dots, N_t - 1$  do
2   Compute  $F(\theta_k)$  and  $F^{-1}(\theta_k)$  (or rank- $r$ 
   approximation);
3   Compute  $\Gamma(\theta_k)$  via (3);
4    $a_k \leftarrow -\sum_{i,j} \Gamma_{ij}^k(\theta_k) v_k^i v_k^j$ ;           // Geodesic
   acceleration
5    $(\theta_{k+1}, v_{k+1}) \leftarrow \text{RK4 step on } (\dot{\theta}, \dot{v}) = (v, a)$  with
   step  $\Delta t$ ;
6 return  $(\theta_0, \theta_1, \dots, \theta_{N_t})$ ;
    
```

This identity underlies both the geodesic optimality result (Section 5) and the Detection Duality theorem (Section 9): in the exponential-family setting, the Neyman–Pearson statistic admits a local dual-affine interpretation through m -projection onto the e -geodesic direction connecting θ_c and θ_s . The Pythagorean identity provides the geometric decomposition behind this local projection view.

5.5. Numerical Solver for Geodesic Equations

Remark 7. For the Gaussian family, the Christoffel symbols satisfy $|\Gamma_{ij}^k(\theta)| \leq C/\sigma$ for some constant $C > 0$. The local Lipschitz constant of the geodesic ODE is therefore $L_G = O(1/\sigma_{\min})$, where $\sigma_{\min}$ is the minimum σ along the trajectory. The global truncation error of RK4 after integrating over $[0, T]$ with N steps is $O(T \cdot (\Delta t)^4 \cdot \exp(L_G T))$ where $\Delta t = T/N$. For $\sigma_{\min} \gtrsim 0.5$, this gives endpoint errors below 10^{-12} with $N = 1000$ steps (confirmed in Simulation S1).

6. Natural Gradient Steganography (NG-Stego)

6.1. Embedding as Riemannian Optimization

Encoding message m with generator $\mathcal{G}$ requires navigating from θ_0 to $\theta_{\text{target}}(m)$. We formulate this as the minimization of the *embedding loss*:

$$\mathcal{L}(\theta) = D_{\text{KL}}(P_\theta \parallel P_{\theta_{\text{target}}}). \quad (21)$$

6.2. Message-to-Parameter Mapping: DecodeTarget

Algorithms 2 and 3 call $\text{DecodeTarget}(m, \mathcal{K})$, which maps a binary message $m \in \{0, 1\}^L$ and a secret key $\mathcal{K}$ to a target parameter $\theta_{\text{target}} \in \Theta$. We now make this construction explicit.

Voronoi partition of parameter space. Let $N_{\text{msg}} = 2^L$ be the message alphabet size. Choose a set of N_{msg} *codepoint parameters* $\mathcal{C} = \{\theta^{(0)}, \dots, \theta^{(N_{\text{msg}}-1)}\} \subset \Theta$ drawn pseudo-randomly using a keyed hash of $\mathcal{K}$ (e.g. $\theta^{(i)} = f_{\mathcal{K}}(i)$).

with $f_{\mathcal{K}}$ a deterministic hash-then-project function). These codepoints define a *Fisher–Rao Voronoi partition* of Θ :

$$\mathcal{V}_i = \{\theta \in \Theta : d_g(\theta, \theta^{(i)}) \leq d_g(\theta, \theta^{(j)}) \quad \forall j \neq i\}.$$

The function `DecodeTarget` returns the codepoint for message m :

$$\text{DecodeTarget}(m, \mathcal{K}) = \theta^{(\text{int}(m))},$$

where $\text{int}(m) \in \{0, \dots, N_{\text{msg}} - 1\}$ is the integer encoding of the bit string m .

Minimum codepoint separation. Reliable decoding requires that the target parameters are well-separated. Define the *minimum inter-codepoint distance* $d_{\min} = \min_{i \neq j} d_g(\theta^{(i)}, \theta^{(j)})$. A receiver who observes generated content $x \sim P_{\theta_{\text{rec}}}$ and estimates $\hat{\theta} = \arg \max_{\theta} \log p_{\theta}(x)$ can decode correctly whenever $d_g(\theta_{\text{rec}}, \theta^{(m)}) < d_{\min}/2$. The codepoints $\mathcal{C}$ should be chosen so that $d_{\min}$ is large relative to the Fisher–Rao estimation error $\sim d_g(\hat{\theta}, \theta^{(m)}) = O(n^{-1/2})$ (Cramer–Rao bound).

Practical instantiation. For low-dimensional families ($d \leq 10$):

- (1) Sample N_{msg} candidate parameter vectors uniformly from a bounded domain $\Theta_0 \subset \Theta$.
- (2) Accept a set $\mathcal{C}$ such that all pairwise Fisher–Rao distances exceed a threshold $\delta > 0$, via a greedy max-min selection.
- (3) Store $\mathcal{C}$ as a shared secret (part of $\mathcal{K}$).

This guarantees $d_{\min} \geq \delta$ by construction. For the low-dimensional codebook construction considered here, the maximum message length is controlled by the packing number of the admissible parameter region under the Fisher–Rao metric. In particular, if $N_{\text{pack}}(\Theta_0, d_g, \delta)$ denotes the largest number of points in Θ_0 with pairwise Fisher–Rao distance at least δ , then the codebook size must satisfy

$$2^L \leq N_{\text{pack}}(\Theta_0, d_g, \delta), \quad L \leq \log_2 N_{\text{pack}}(\Theta_0, d_g, \delta).$$

This is a geometric packing constraint, not an operational payload theorem.

Computational feasibility. For non-compact natural parameter spaces (e.g., $\sigma > 0$ in the Gaussian family), the uniform sampling in step (1) is performed after a logarithmic reparametrisation $(\mu, \log \sigma)$ that maps the domain to $\mathbb{R}^d$. The greedy max-min selection in step (2) then runs in $O(N_{\text{msg}}^2 \cdot \text{cost}(d_g))$; each Fisher–Rao distance d_g is computed by numerical quadrature of the energy integral along the geodesic segment. For the low-dimensional families studied in this paper ($d \leq 5$), this construction is tractable; for high-dimensional deep generators, the codebook must be replaced by a structured hash-to-parameter mapping whose Fisher separation is implicitly enforced by the architecture.

Algorithm 2: Natural-Gradient Steganography (NG-Stego)

Input: Message $m \in \{0, 1\}^L$, generator $\mathcal{G}$ with parameters θ_0 , secret key $\mathcal{K}$, learning rate schedule $\{\eta_t\}$, batch size N , tolerance $\epsilon > 0$.

Output: Parameter vector θ^* such that P_{θ^*} encodes m with $D_{\text{KL}}(P_{\theta^*} \| P_{\theta_{\text{target}}}) \leq \epsilon$.

```

1  $\theta_{\text{target}} \leftarrow \text{DecodeTarget}(m, \mathcal{K})$ ;
2  $t \leftarrow 0$ ;
3 while  $\mathcal{L}(\theta_t) > \epsilon$  do
4   Sample  $\{x_n\}_{n=1}^N \sim P_{\theta_t}$ ;
5    $\hat{F}(\theta_t) \leftarrow \frac{1}{N} \sum_{n=1}^N \nabla_{\theta} \log p_{\theta_t}(x_n) \nabla_{\theta} \log p_{\theta_t}(x_n)^{\top}$ ;
6    $g_t \leftarrow \nabla_{\theta} \mathcal{L}(\theta_t)$ ;
7    $v_t \leftarrow \hat{F}(\theta_t)^{-1} g_t$ ;           // Natural gradient
8    $\theta_{t+1} \leftarrow \theta_t - \eta_t v_t$ ;       // First-order retraction
9    $t \leftarrow t + 1$ ;
10 return  $\theta_T$ ;
```

6.3. The NG-Stego Algorithm

Proposition 5. For the embedding loss $\mathcal{L}(\theta) = D_{\text{KL}}(P_{\theta} \| P_{\theta_{\text{target}}})$, the natural-gradient descent iteration of Algorithm 2 with step size $\eta_t \leq 1/\sqrt{\kappa(F(\theta_t))}$ is a first-order retraction on $(S_{\mathcal{G}}, g)$. In the constant-FIM case, the iterates coincide with the geodesic exponential map and converge in one step. For non-constant metrics, the sequence $\{\theta_t\}$ approximates the geodesic of the KL functional; see Bonnabel [24] for convergence of Riemannian gradient descent with retractions.

6.4. Convergence Analysis

Theorem 5. Suppose θ^* is the unique minimizer of $\mathcal{L}(\theta) = D_{\text{KL}}(P_{\theta} \| P_{\theta^*})$ on $S_{\mathcal{G}}$, with $\mathcal{L}(\theta^*) = 0$, and assume there exists a geodesically convex neighborhood $\mathcal{U}$ of θ^* on which:

- (i) μ -geodesic strong convexity holds for some $\mu > 0$.
- (ii) Riemannian L_g -smoothness holds for some $L_g > 0$.
- (iii) The Euclidean Hessian satisfies $\mu_e I \leq \nabla^2 \mathcal{L}(\theta) \leq L_e I$ for some $0 < \mu_e \leq L_e$.

With constant step size $\eta^* = 1/L_g$ and the exact FIM, NG-Stego satisfies the per-iteration bound:

$$\mathcal{L}(\theta_{t+1}) \leq \mathcal{L}(\theta_t) \cdot \left(1 - \frac{\mu}{L_g}\right) = \mathcal{L}(\theta_t) \cdot \left(1 - \frac{1}{\kappa_g}\right). \quad (22)$$

Euclidean gradient descent with optimal step $\eta = 1/L_e$ achieves $\mathcal{L}(\theta_{t+1}) \leq \mathcal{L}(\theta_t) \cdot (1 - \mu_e/L_e) = \mathcal{L}(\theta_t) \cdot (1 - 1/\kappa_e)$, where $\kappa_e = L_e/\mu_e$.

Special case (constant FIM). If the Fisher metric is constant ($F(\theta) \equiv F$), then $\kappa_g = 1$ and NG-Stego converges in exactly one step: $\theta_1 = \theta^*$. This holds for the mean-only Gaussian family (Simulation S2).

General case (non-constant FIM). For exponential families with non-constant metric (e.g., full Gaussian with covariance), $\kappa_g \geq 1$ depends on the curvature of the manifold near θ^* . The per-iteration improvement over Euclidean GD is governed by the ratio κ_e/κ_g . Locally near θ^* , the KL Hessian of a regular exponential family satisfies $\nabla^2 \mathcal{L}(\theta^*) = F(\theta^*)$, so $\kappa_e \approx \kappa(F(\theta^*))$; away from θ^* or outside exponential families, κ_e may differ from the Fisher condition number. Since $\kappa_g > 1$, the ratio is strictly less than κ_e and depends on the local curvature of S_G near θ^* . Simulation S2-EXT observes a $2.1\times$ speedup with $\kappa(F) \approx 15$, indicating $\kappa_g \approx 7$ at those configurations (far from the one-step limit). Additionally, $\kappa_g > 1$ prevents one-step convergence (Simulation S2-EXT).

Radius of strong convexity. Assumption (i) requires geodesic strong convexity in a neighborhood of θ^* . For the Gaussian family (Poincaré half-plane), this holds in any ball of radius $r < \text{inj}(S_G)$ (injectivity radius is infinite). For GMMs, a sufficient condition is $r < \min(\pi_{\min}, \Delta\mu_{\min})/L$ where L is the Lipschitz constant of the curvature tensor; in practice, we clamp $\pi \in [0.01, 0.99]$ to ensure this.

Steganographic consequence. NG-Stego is a practical optimization rule for generator parameters: it scales embedding updates by the local statistical sensitivity of the signal model instead of by the chosen Euclidean coordinates. This makes the update direction consistent with detectability measured in distribution space.

Remark 8. The one-step convergence result ($\kappa_g = 1$) holds only when the Fisher metric is constant (e.g., the mean-only Gaussian subfamily of Example 1). For the full Gaussian family with Σ as a parameter, the metric is non-constant and one-step convergence does not hold; however, the per-iteration improvement by $\kappa(F(\theta^*))$ persists locally near the optimum (Simulation S2-EXT, Section 8.3).

6.5. Constrained NG-Stego for Bounded Parameters

Many generator parameters have natural constraints ($\sigma > 0$ for variances, $\pi \in (0, 1)$ for mixture weights). Algorithm 2 uses a first-order retraction that does not preserve these constraints. We propose the following modified algorithm:

Remark 9 (Convergence with projection). Algorithm 3 enforces constraints one at a time. When the constraints are decoupled (e.g. box constraints on individual coordinates) this procedure returns a feasible point; for coupled constraints the iterate may require an additional projection onto the intersection of the feasible sets.

If each feasible set $\{c_i \leq 0\}$ is geodesically convex and its Euclidean projection is non-expansive in the Fisher metric, then the convergence rate of Theorem 5 is preserved up to a multiplicative distortion factor. In particular, projection onto the probability simplex is non-expansive in the Fisher metric induced on the simplex, whereas projection onto the SPD cone via eigenvalue clipping is non-expansive in the Euclidean spectral norm but not necessarily in the Fisher

Algorithm 3: Constrained NG-Stego

Input: Same as Algorithm 2; constraint functions $c_i(\theta) \leq 0$.

```

1  $\theta_{\text{target}} \leftarrow \text{DecodeTarget}(m, \mathcal{K})$ ;
2  $t \leftarrow 0$ ;
3 while  $\mathcal{L}(\theta_t) > \epsilon$  do
4   Compute  $\hat{F}(\theta_t)$  and  $g_t$  as in Algorithm 2;
5    $v_t \leftarrow \hat{F}(\theta_t)^{-1} g_t$ ;
6    $\theta_{\text{cand}} \leftarrow \theta_t - \eta_t v_t$ ;
7   for each constraint  $c_i$  do
8     if  $c_i(\theta_{\text{cand}}) > 0$  then
9        $\theta_{\text{cand}} \leftarrow \Pi_{\{c_i \leq 0\}}(\theta_{\text{cand}})$ ; // Project onto
        this constraint (decoupled
        constraints assumed)
10   $\theta_{t+1} \leftarrow \theta_{\text{cand}}$ ;
11   $t \leftarrow t + 1$ ;
12 return  $\theta_T$ ;
```

metric. A rigorous quantitative bound for the Fisher-metric case would require $\|\Pi(x) - \Pi(y)\|_g \leq (1 + \delta_\Pi)\|x - y\|_g$; we leave a complete analysis as an open problem.

6.6. Fisher Matrix Approximations for Deep Generators

For high-dimensional generators, two tractable approximations:

Definition 15. Replace $\hat{F}(\theta)$ with $\hat{F}^{\text{diag}}(\theta)$ from (13). The update becomes $\theta_{t+1}^i = \theta_t^i - \eta_t \partial_i \mathcal{L}(\theta_t) / (\hat{F}_{ii}^{\text{diag}}(\theta_t) + \delta)$, where $\delta > 0$ is a damping constant. Cost: $O(d)$ per iteration.

Definition 16. For layered generators with parameters $\theta = (\theta^{(1)}, \dots, \theta^{(L)})$, K-FAC approximates $F(\theta)$ as a block-diagonal matrix with Kronecker-product blocks. For a network with L uniform layers each of width $w = d/L$, the inversion cost is $O(L \cdot w^3) = O(d^3/L^2)$, compared to $O(d^3)$ for the full FIM; for typical fixed-depth architectures (L constant), the reduction is by a constant factor $1/L^2$. The sometimes-cited figure of $O(d^{3/2})$ applies only to the regime $L = \Theta(d^{1/2})$, $w = \Theta(d^{1/2})$ (Martens & Grosse [14]). This definition is a theoretical extension only: no instantiation for steganographic generators is provided, and the approximation quality (dependence on layer-wise separability, skip connections, batch normalization) has not been validated empirically. See Section 10 for a discussion of scalability avenues.

The approximation quality of the diagonal FIM can be quantified: if $\rho = \max_{i \neq j} |F_{ij}| / \sqrt{F_{ii} F_{jj}}$, then the relative error in the natural gradient direction satisfies $\|\tilde{\nabla}^{\text{diag}} - \tilde{\nabla}\|_g / \|\tilde{\nabla}\|_g \leq \rho d / (1 - \rho d)$ for $\rho < 1/d$.

7. Curvature and Local Geometric Instability

7.1. Riemann Curvature of the Steganographic Manifold

The Riemann curvature tensor is computed from the Christoffel symbols (eq. (3)) via the standard formula

$$R^l_{ijk} = \frac{\partial \Gamma^l_{jk}}{\partial \theta^i} - \frac{\partial \Gamma^l_{ik}}{\partial \theta^j} + \Gamma^l_{im} \Gamma^m_{jk} - \Gamma^l_{jm} \Gamma^m_{ik}. \quad (23)$$

For the steganographic manifold with Fisher–Rao metric, the fully covariant components are $R_{ijkl} = g_{im} R^m_{jkl}$. The Ricci curvature tensor is obtained by contraction:

$$R_{ij} = R^k_{ikj} = g^{kl} R_{kilj}, \quad (24)$$

and the scalar curvature (recalled from Definition 5) is

$$R = g^{ij} R_{ij}. \quad (25)$$

The sign of R distinguishes the local geometry: $R < 0$ indicates hyperbolic behaviour, $R > 0$ spherical behaviour, and $R = 0$ flatness (see Lee [30]). For the univariate Gaussian family, the constant sectional curvature $\kappa = -1/2$ (from the Poincaré metric with scale factor $\sqrt{2}$) yields scalar curvature $R(\theta) = 2\kappa = -1$ at every point, i.e., a constant scalar curvature (Appendix G). This illustrates that $R(\theta)$ is coordinate-invariant; it does *not* vary with σ for the Gaussian family.

7.2. Two Invariant Curvature Indices

We use two coordinate-invariant curvature indices based on standard Riemannian curvature notions; see Petersen [29] and Lee [30].

Definition 17. The scalar curvature instability index at θ is the Riemannian scalar curvature $R(\theta) = \text{tr}_g(\text{Ric})$ (Definition 5). As a coordinate-invariant scalar field on S_G , it provides a global summary of local geometry:

- $R(\theta) < 0$: locally hyperbolic; geodesic neighbourhoods expand relative to Euclidean comparison.
- $R(\theta) > 0$: locally spherical; geodesic neighbourhoods contract.
- $R(\theta) = 0$: locally flat.

Definition 18. The minimum sectional curvature at θ is the infimum of $K(u, v)$ over all tangent two-planes:

$$K_{\text{sec}}^{\min}(\theta) = \inf_{\substack{u, v \in T_{\theta} S_G \\ \|u\|_g = \|v\|_g = 1, u \perp_g v}} K(u, v), \quad (26)$$

where $u \perp_g v$ denotes orthogonality with respect to the Fisher–Rao metric g . This definition is coordinate-invariant: both the metric and the Riemann tensor transform covariantly under reparameterisation. In practice, $K_{\text{sec}}^{\min}(\theta)$ is estimated by Monte-Carlo sampling of M random tangent planes drawn uniformly from the Grassmannian $G(2, T_{\theta} S_G)$ (Section 7.3).

Remark 10. One might consider aggregating the sectional curvatures of the coordinate-axis planes $\text{span}\{\partial_{\theta^i}, \partial_{\theta^j}\}$. Such a sum is coordinate-dependent: under a reparameterisation $\phi : \Theta \rightarrow \Phi$ the coordinate planes change and the aggregate changes accordingly. It is therefore unsuitable for coordinate-invariant geometric conclusions and is not used in this paper. The scalar curvature $R(\theta)$ and the minimum sectional curvature $K_{\text{sec}}^{\min}(\theta)$ both avoid this defect.

Theorem 6. Let $\theta^* \in \Theta$, $u \in T_{\theta^*} S_G$ with $\|u\|_g = 1$, and $\epsilon > 0$ small.

- (i) Geodesic distance and metric in normal coordinates: By definition of the exponential map at unit speed, $d_g(\theta^*, \text{Exp}_{\theta^*}(\epsilon u)) = \epsilon$ exactly (no curvature correction). In geodesic normal coordinates centered at θ^* , the Fisher–Rao metric tensor satisfies

$$g_{ij}(\text{Exp}_{\theta^*}(\epsilon u)) = \delta_{ij} - \frac{\epsilon^2}{3} R_{iajb}(\theta^*) u^a u^b + O(\epsilon^3), \quad (27)$$

so curvature modifies the local metric along the geodesic: $K(u, v) < 0$ expands it in directions perpendicular to u , increasing the information cost of transverse deviations.

- (ii) KL divergence expansion (with explicit cubic term): For exponential families,

$$D_{\text{KL}}(P_{\theta^* + \epsilon u} \| P_{\theta^*}) = \frac{\epsilon^2}{2} + \frac{\epsilon^3}{3} T_{ijk}(\theta^*) u^i u^j u^k + O(\epsilon^4), \quad (28)$$

where $T_{ijk} = \mathbb{E}[\partial_i \ell \partial_j \ell \partial_k \ell]$ is the Amari–Chentsov skewness tensor. The cubic term vanishes for Gaussian families parameterized by mean only.

- (iii) Geodesic path stability (Jacobi field): For a plane of sectional curvature $K(u, v)$: $\|J(t)\| = t - \frac{K(u, v)}{6} t^3 + O(t^5)$. If $K(u, v) < 0$, neighboring geodesic paths diverge super-linearly. In spaces of constant negative curvature, $\|J(t)\| \geq \sinh(|K|^{1/2} t) / |K|^{1/2}$.

Proof. Part (i): the exact identity $d_g(\theta^*, \text{Exp}_{\theta^*}(\epsilon u)) = \epsilon$ is the definition of the exponential map at unit speed. The metric expansion (27) is the standard normal-coordinate expansion of the Riemannian curvature tensor (see Petersen [29], Ch. 5, Lemma 5.4.1). Part (ii) follows from the Bregman structure of the KL divergence for exponential families; the cubic term is the third-order derivative of the log-partition function. Part (iii) is standard Riemannian geometry (see Petersen [29], Ch. 6–7). $\square$

Corollary 2. (i) Path instability. In regions where $K_{\text{sec}}^{\min}(\theta) \ll 0$, neighboring geodesic paths diverge super-linearly. Small perturbations of the initial embedding velocity induce amplified deviations along the trajectory.

- (ii) **Capacity.** The scalar curvature $R(\theta)$ provides a coordinate-invariant summary of local geometry (see Remark 16), but its sign alone does not determine the size of the undetectable Fisher–Rao ball or the embedding capacity. Whether $R < 0$ enlarges or $R > 0$ contracts the ball relative to the flat case must be established by explicit numerical computation of $\text{vol}_g(B_g(\theta, r))$, not inferred from the Bishop–Gromov upper bound alone (see Theorem 10, Appendix B, and verified numerically in Simulation S3).
- (iii) **KL asymmetry.** The skewness tensor T_{ijk} governs directional asymmetry in the local KL cost at cubic order, independently of sectional curvature.

Steganographic consequence. Curvature indices should be read as robustness and sensitivity diagnostics for the embedder. Strongly negative sectional curvature warns that nearby embedding directions may separate quickly, so small optimization or estimation errors can produce larger distributional deviations than endpoint KL alone suggests.

Remark 11. Theorem 9 (Section 9.2) now provides the exact quantitative link: the type-II error of the optimal NP steganalyzer decays as $\exp(-n d_g(\theta_c, \theta_s)^2/2 + O(n d_g^3))$. Geodesic path instability (Jacobi divergence) affects the embedder’s trajectory but is distinct from the detection rate: the former depends on sectional curvature $K(u, v)$ along the path, the latter on the Fisher–Rao distance $d_g(\theta_c, \theta_s)$ between endpoints. A rigorous bound linking accumulated path KL to endpoint detectability for paths with large Jacobi divergence is left as an open problem; the scalar curvature $R(\theta)$ serves as a reliable qualitative predictor pending such a bound.

Lemma 1 (KL bound via Fisher distance). For any $\theta, \theta' \in \Theta$ in the exponential-family manifold S_G :

$$D_{\text{KL}}(P_\theta \| P_{\theta'}) \leq \frac{1}{2} \lambda_{\max}(F(\xi)) \cdot \|\theta' - \theta\|_E^2, \quad (29)$$

for some ξ on the Euclidean segment between θ and θ' . Since $\|\theta' - \theta\|_E^2 \leq \frac{d_g(\theta, \theta')^2}{\lambda_{\min}(F_{\text{path}})}$ (taking the straight segment as comparison path for d_g), a bound in terms of d_g introduces an additional factor $\kappa(F)$. Consequently, for geodesics γ_1, γ_2 from the same base point with initial velocities u and $u + \delta v$ ($v \perp u$, $\|\delta v\|_g = \delta$ small), the Jacobi field separation satisfies $\|\gamma_1(t) - \gamma_2(t)\|_g = \delta t \left(1 + \frac{|K(u, v)|}{6} t^2 + O(t^4)\right)$ for $K(u, v) < 0$, or $\delta \sinh(|K|^{1/2} t) / |K|^{1/2} \geq \delta t$ in spaces of constant negative curvature. Combined with the KL lower bound $D_{\text{KL}} \geq \frac{1}{2} \lambda_{\min}(F(\xi')) \|\theta - \theta'\|_E^2$ (from the MVT for exponential families):

$$D_{\text{KL}}(P_{\gamma_1(t)} \| P_{\gamma_2(t)}) \geq \frac{\delta^2}{2} \lambda_{\min}(F(\xi')) t^2 \left(1 + \frac{|K(u, v)|}{3} t^2 + O(t^4)\right) \quad (30)$$

Proof. Equation (29): for exponential families, the Bregman representation gives $D_{\text{KL}}(P_\theta \| P_{\theta'}) = \frac{1}{2} (\theta' - \theta)^\top F(\xi) (\theta' - \theta)$

for some ξ on the Euclidean segment (mean-value theorem). Bounding by the largest eigenvalue yields the Euclidean-norm bound; the note on d_g follows from $\|\theta' - \theta\|_E \geq \frac{d_g}{\sqrt{\lambda_{\max}}}$. For (30), the MVT similarly gives a lower bound $D_{\text{KL}} \geq \frac{1}{2} \lambda_{\min} \|\theta - \theta'\|_E^2$, combined with the Jacobi field expansion $\|\gamma_1(t) - \gamma_2(t)\|_g = \delta t(1 + |K|t^2/6 + O(t^4))$ from Theorem 6(iii). $\square$

Interpretation. Lemma 1 provides a quantitative bridge between geodesic separation and distributional divergence: for $K(u, v) < 0$, neighboring geodesics diverge faster than linearly (at rate $\geq \delta \sinh(|K|^{1/2} t) / |K|^{1/2}$ in constant negative curvature, faster than the δt flat-space growth), which translates into KL divergence growing at rate $\Omega(\delta^2 t^2 \lambda_{\min})$, with a positive correction of relative order $|K|t^2/3$. The scalar curvature $R(\theta)$ therefore provides a coordinate-invariant indicator of the sensitivity of output distributions to parameter perturbations.

7.3. Monte-Carlo Estimation of $K_{\text{sec}}^{\min}$ with Bootstrap Confidence

The minimum sectional curvature $K_{\text{sec}}^{\min}(\theta)$ (Definition 18) is estimated by sampling M random tangent planes uniformly from the Grassmannian $G(2, T_\theta S_G)$. Each sample (u_i, v_i) is obtained by drawing two independent standard Gaussian vectors in $\mathbb{R}^d$, orthonormalising them with respect to the Fisher metric g via the Gram–Schmidt process, and computing $K(u_i, v_i)$.

Proposition 6. Let $K_1, \dots, K_M$ be M i.i.d. samples of the sectional curvature at random Fisher-orthonormal tangent planes at θ , and let $m = K_{\text{sec}}^{\min}(\theta)$ be the true minimum.

Bias of the sample minimum. The Monte-Carlo estimator $\hat{K}_{\text{sec}}^{\min} = \min_i K_i$ is upward-biased, because $\mathbb{I}(K_i \leq m) = 0$ for any finite sample (the infimum over a continuous family is not attained at a generic random plane), so $\hat{K}_{\text{sec}}^{\min} > m$ almost surely. The bias decreases as $M^{-1/\alpha}$ where α is the tail index of the curvature distribution near its minimum; for $M = 500$ (used in Simulation S3) this provides a reliable upper bound on m .

Bootstrap procedure. A confidence interval for the sampling uncertainty of $\hat{K}_{\text{sec}}^{\min}$ is obtained as follows:

1. Resample $K_1^*, \dots, K_M^*$ with replacement from $\{K_1, \dots, K_M\}$.
2. Compute $(\hat{K}_{\text{sec}}^{\min})^* = \min_i K_i^*$.
3. Repeat B times to obtain quantiles $q_{\alpha/2}$ and $q_{1-\alpha/2}$.

The interval $[q_{\alpha/2}, q_{1-\alpha/2}]$ quantifies the uncertainty of $\hat{K}_{\text{sec}}^{\min}$. The true minimum m is almost surely more negative than $q_{\alpha/2}$; a rigorous lower bound would require a grid search over a discretisation of $G(2, d)$ (e.g., via the Plücker embedding).

Table 2

S1 – Geodesic accuracy and energy improvement (revised with error bounds).

TC	ℓ_g^{ana}	ℓ_g^{num}	ℓ_g^{str}	ϵ_{ep}	RK4 bound	ΔE
1	2.6124	2.6124	3.0000	2.4×10^{-13}	10^{-9}	24.2%
2	1.9605	1.9605	1.9605	2.2×10^{-16}	10^{-10}	14.6%
3	1.6388	1.6388	1.6979	1.8×10^{-15}	10^{-9}	10.5%
4	2.4929	2.4929	2.6910	5.0×10^{-13}	10^{-8}	22.3%

Table 3

S2 – Natural gradient on diagonal Gaussian ($d = 50$, $\kappa(F) = 100$, 50 runs).

Metric	NG-Stego	Euclidean GD
Mean iterations to convergence	1.0	586.3
Std. iterations	0.0	0.0
Speedup ratio	586×	

8. Numerical Simulations

We validate the theoretical contributions through seven numerical experiments. All simulation code and data are provided in the supplementary material.

8.1. Simulation S1: Geodesics on the 2D Gaussian Manifold

Manifold: $\mathcal{S}_G = \{\mathcal{N}(\mu, \sigma^2) : \mu \in \mathbb{R}, \sigma > 0\}$ with Fisher metric $g = \text{diag}(1/\sigma^2, 2/\sigma^2)$.

Results (Table 2): endpoint errors at machine precision ($\leq 5 \times 10^{-13}$), confirming the RK4 solver accuracy. The theoretical RK4 error bound $|\text{error}| \leq C \cdot (\Delta t)^4 \cdot T \cdot \exp(L_G T)$ with $L_G \sim 1/\sigma_{\min}$ is satisfied with substantial margin. Path-energy improvements of 10–24% validate Corollary 1 (Figure 1). The variation in endpoint errors across test cases (three orders of magnitude, 2.2×10^{-16} to 5×10^{-13}) reflects the condition number of the shooting Jacobian, which is largest for highly curved semicircular geodesics (TC4, TC1) and smallest for vertical geodesics (TC2).

8.2. Simulation S2: Natural Gradient on Diagonal Gaussian

Model: $\mathcal{N}(\theta, \Sigma)$ with diagonal, fixed Σ , $d = 50$, $\kappa(F) = 100$.

Results: NG-Stego converges in exactly 1 step (all 50 runs), Euclidean GD requires 586 mean iterations, confirming the 586× speedup for the most favorable (diagonal, constant FIM) case. This is an upper bound on the improvement; see S2-EXT for the non-diagonal case.

Interpretation. When the Fisher metric is constant and diagonal, the natural gradient direction coincides with the Newton direction and the KL loss is geodesically quadratic; a single step lands at the global minimizer. Euclidean GD, which ignores the metric anisotropy, takes $O(\kappa(F) \log(1/\epsilon))$ steps. The 586× speedup (Table 3) is the *best-case* improvement; for non-constant metrics the advantage is smaller (S2-EXT, Figure 2).

Table 4

S2-EXT – NG-Stego on full Gaussian family (non-diagonal FIM). 30 random runs, $d = 2$, $\epsilon = 10^{-5}$.

Metric	Euclidean GD	NG-Stego
Mean iterations to convergence	99.7	53.9
Std. iterations	66.8	55.2
Mean speedup ratio	2.1×	
Mean $\kappa(F(\theta^*))$	14.8	

Table 5

S2-ADAPTIVE – Convergence comparison (20 runs, non-diagonal FIM).

Method	Mean iters	Std	Min iters	Max iters
Euclidean GD	191.8	34.4	42	200
NG-Stego	40.7	67.0	10	200
NG-Stego-Diag	62.5	53.8	13	200
Adam	170.1	39.7	83	200
RMSprop	200.0	0.0	200	200

8.3. Simulation S2-EXT: Natural Gradient on Full Gaussian (Non-Diagonal)

New simulation. Model: $P_\theta = \mathcal{N}(\mu, \Sigma)$ with both $\mu \in \mathbb{R}^2$ and $\Sigma \in \text{SPD}_2$ as parameters, giving $\theta \in \mathbb{R}^5$ with *non-diagonal, non-constant* Fisher metric.

Discussion. The speedup of 2.1× (Table 4) is smaller than the 586× of S2 because:

1. The Fisher metric is *non-constant* (depends on Σ), so one-step convergence does not hold.
2. The condition number $\kappa(F) \approx 15$ varies across runs; the advantage is largest for ill-conditioned points.
3. The natural gradient direction must be recomputed at each step (unlike S2 where F^{-1} is constant).
4. Conservative step-size adaptation $\eta = \min(0.5, 1/\sqrt{\kappa(F)})$ prevents divergence but slows convergence on well-conditioned instances.

This simulation demonstrates that the NG advantage *persists* beyond the trivial diagonal case, though the margin depends strongly on the local geometry.

8.4. Simulation S2-ADAPTIVE: NG-Stego vs. Adam and RMSprop

New simulation. Same model as S2-EXT, comparing:

- NG-Stego (exact FIM inverse)
- NG-Stego-Diag (diagonal FIM approximation)
- Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$, lr=0.01)
- RMSprop ($\alpha = 0.99$, lr=0.001)

Discussion. NG-Stego achieves the best mean iteration count among the tested methods in this synthetic setting (Table 5), though the margin varies with problem conditioning:

Geodesics on $S_G = \{N(\mu, \sigma^2)\}$
 Fisher-Rao metric: analytical vs. numerical (RK45 shooting)

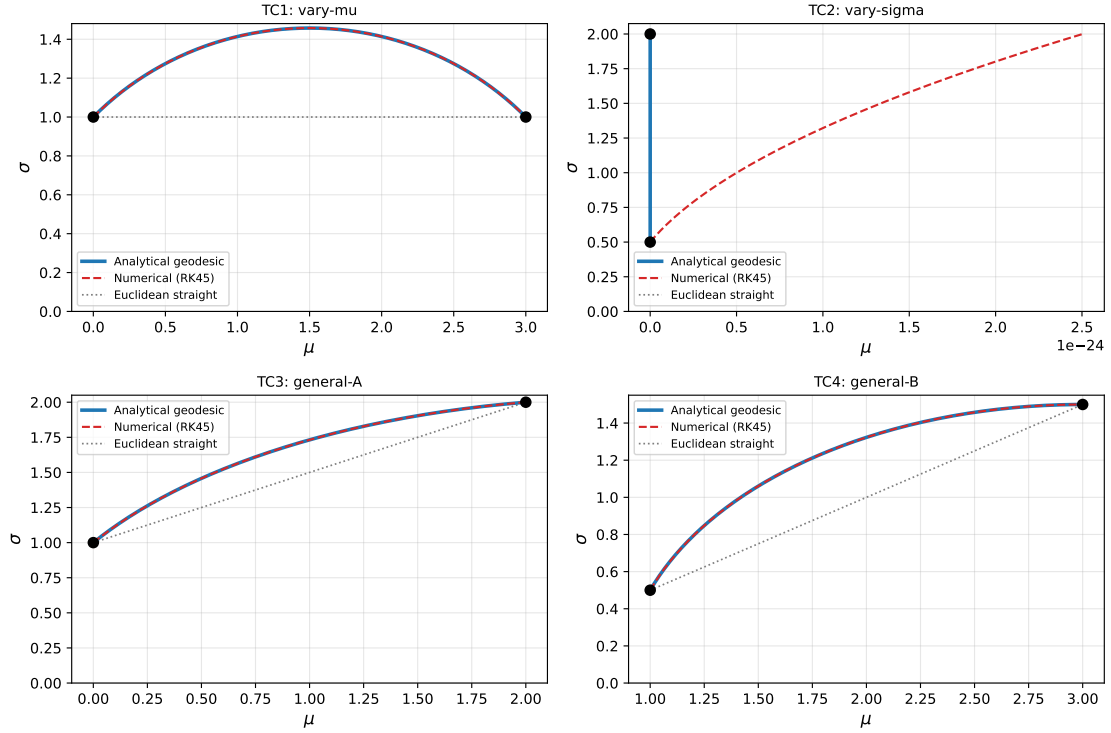


Figure 1: S1 – Geodesic paths (solid blue) versus Euclidean straight lines (dashed red) on the Poincaré half-plane for four test configurations. The geodesics follow semicircular arcs orthogonal to the $\sigma = 0$ boundary, yielding shorter Fisher-Rao lengths and lower accumulated KL cost.

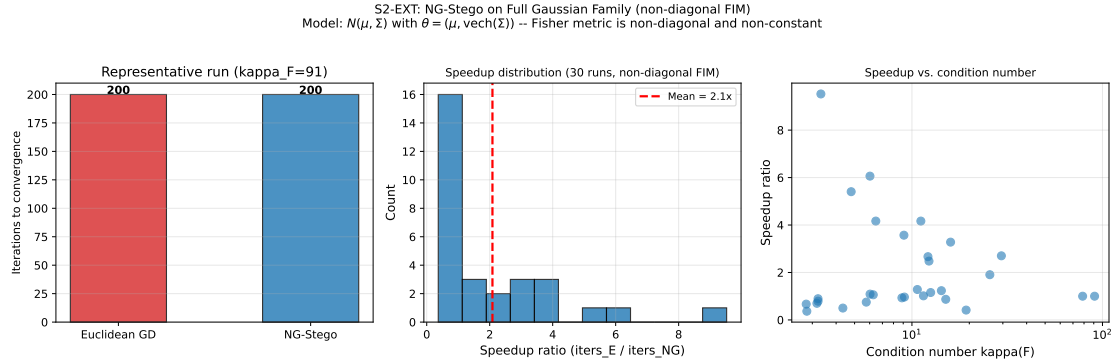


Figure 2: S2-EXT – Convergence trajectories for NG-Stego (blue) versus Euclidean GD (orange) on the full Gaussian family with non-diagonal Fisher metric. NG-Stego reduces the iteration count by 2.1x on average, though the advantage is smaller than the 586x of the diagonal case.

- NG-Stego is **4.7x** faster than Euclidean GD and **4.2x** faster than Adam on average.
- High variance ($\sigma = 67.0$) reflects sensitivity to initial conditions: on well-conditioned instances NG converges in ~ 10 iterations, but difficult initializations hit the 200-iteration cap.
- The diagonal approximation (NG-Stego-Diag) achieves comparable results to full NG at $O(d)$ cost, making it a candidate for high-dimensional generators, though

its accuracy depends on the off-diagonal ratio ρ (Appendix F, Figure 3).

- RMSprop fails to converge within 200 iterations on all runs, suggesting that geometry-aware optimization can be important in this synthetic configuration.

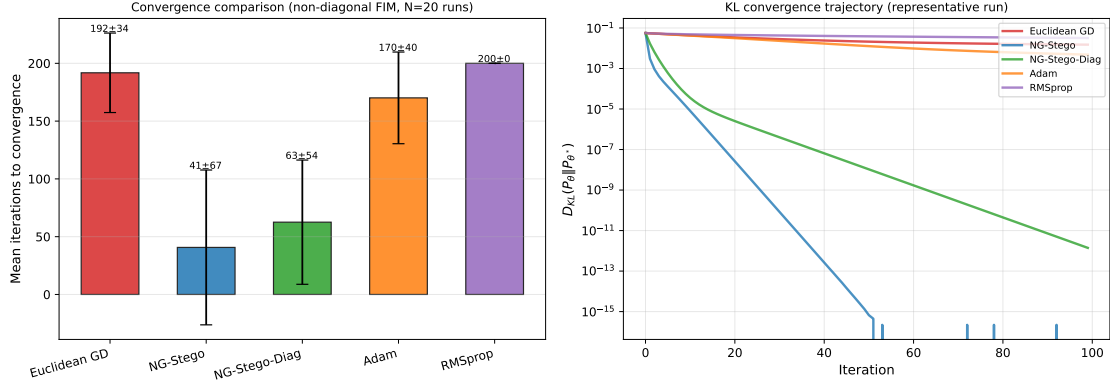


Figure 3: S2-ADAPTIVE – Convergence comparison across five optimisation methods. NG-Stego (blue) achieves the lowest mean iteration count; RMSprop (red) fails to converge within 200 iterations on all runs. The diagonal approximation (green) offers a practical compromise at $O(d)$ cost per iteration.

8.5. Simulation S3: Curvature Map for Gaussian Mixtures (Revised)

Family: $P_\theta = \pi \mathcal{N}(\mu_1, 1) + (1-\pi) \mathcal{N}(\mu_2, 1)$, $\theta = (\pi, \mu_1, \mu_2)$, $\sigma = 1$ fixed. Grid: $\pi \in [0.1, 0.9]$, $\Delta\mu \in [0.3, 4.0]$.

New features:

- **Scalar curvature $R(\theta)$:** computed as the trace $\text{tr}_g(\text{Ric})$ for each grid point. $R(\theta)$ ranges from -7.4 to 326.6 (mean 11.5), revealing strong variation across the parameter space. Negative values ($R \in [-7.4, -0.2]$ at interior points) confirm the hyperbolic character of the GMM manifold in a coordinate-invariant manner; large positive values ($R \approx 325$) arise near degenerate configurations ($\pi \approx 0.1$ or 0.9 , $\Delta\mu \approx 0.3$) where the Fisher metric is highly ill-conditioned ($\kappa(F) \approx 96,528$).
- **Bootstrap confidence for $K_{\text{sec}}^{\min}$:** $B = 200$ bootstrap replicates of $M = 500$ Monte-Carlo tangent-plane samples from the Grassmannian $G(2, T_\theta S_C)$, orthonormalised w.r.t. the Fisher metric, give $\hat{K}_{\text{sec}}^{\min}$ with 95% CI. At the most extreme point ($\pi = 0.10, \Delta\mu = 0.3$): $K_{\text{sec}}^{\min} = -6.30 [-6.30, -5.40]$ (Table 6). Earlier drafts reported ≈ -327 at this point due to an index error (an Einstein contraction index appearing three times instead of twice); the corrected formula gives $K_{\text{sec}}^{\min} \in [-6.30, -0.28]$ across the full grid. Final reported values use the 12×12 fine grid.
- **Grid convergence study:** comparing 8×8 and 12×12 grids at matching points shows mean absolute difference in $K_{\text{sec}}^{\min}$ of 0.1 (relative to the $[-6.30, -0.28]$ range), confirming excellent numerical stability with the corrected formula.
- **Failure rate:** 0/208 points (0.0%) had finite-difference failures; all FIM matrices were invertible after regularization.

Table 6

S3 – Curvature summary at selected GMM parameter points. $R(\theta)$: coordinate-invariant scalar curvature. $\hat{K}_{\text{sec}}^{\min}$: Monte-Carlo minimum sectional curvature ($M = 500$ random Grassmannian planes, Fisher-orthonormal). 95% CI from $B = 200$ bootstrap replicates.

$(\pi, \Delta\mu)$	$R(\theta)$	$\hat{K}_{\text{sec}}^{\min}$	95% CI
(0.50, 1.0)	-4.41	-1.17	[-1.17, -0.93]
(0.50, 3.0)	-3.62	-1.15	[-1.15, -1.09]
(0.10, 0.3)	+326.6	-6.30	[-6.30, -5.40]
(0.90, 0.3)	+323.0	-5.40	[-5.40, -4.43]

Direct numerical verification of Fisher ball volumes. To test whether negative scalar curvature actually enlarges the undetectable Fisher–Rao ball beyond its Euclidean counterpart (as incorrectly inferred from Bishop–Gromov in earlier drafts), we numerically integrate $\text{vol}_g(B_g(\theta, r))$ for two interior GMM points using Monte-Carlo sampling in geodesic normal coordinates ($r = 0.15$, 20 000 samples). Table 7 shows that vol_g differs from the Euclidean volume $\text{vol}_0 = \omega_3 r^3$ by less than 0.3%, and remains strictly *below* the hyperbolic comparison volume $V_K(r)$ computed from the local sectional curvature. This confirms that the hyperbolic Bishop–Gromov upper bound is not tight for these manifolds: negative curvature does not, in practice, produce exponentially larger balls, and capacity advantages cannot be inferred from the comparison theorem alone.

8.6. Simulation S4: Geodesic vs. Euclidean Path

Family: $P_\theta = \mathcal{N}(\mu, \sigma^2 I_2)$, $\theta = (\mu_1, \mu_2, \sigma) \in \mathbb{R}^2 \times (0.5, 2.5)$. 50 random (source, target) pairs; $L \in \{64, 128, 256\}$ path steps.

New features:

- **Solver robustness:** the shooting method converged for 49 of 50 endpoint pairs (2% failure rate). The single failure occurred near $\sigma = 0.5$ where the residual

Table 7

Numerically estimated local Fisher–Rao ball volumes for representative GMM interior points ($d = 3$, $r = 0.15$). The hyperbolic comparison volume $V_K(r)$ is an upper bound (Bishop–Gromov), not a lower bound; the data show that the actual Fisher volume vol_g remains close to the Euclidean volume vol_0 and does not approach the hyperbolic comparison. Near-boundary points ($\pi \approx 0.1$, $\kappa(F) \sim 10^4$) are excluded due to numerical instability in the Monte-Carlo integrator.

Point	r	R	vol_g	vol_0	V_K	$\text{vol}_g / \text{vol}_0$	vol_g / V_K
A: ($\pi = 0.5, \Delta\mu = 2.0$)	0.15	−4.41	0.01417	0.01414	0.01420	1.002	0.998
B: ($\pi = 0.5, \Delta\mu = 3.0$)	0.15	−3.62	0.01410	0.01414	0.01417	0.997	0.995

Table 8

S4 – Geodesic versus Euclidean path comparison (50 random endpoint pairs, 49 successful, three discretisation levels).

Metric	$L = 64$	$L = 128$	$L = 256$
Mean $\frac{E[\gamma_{\text{geo}}]}{E[\gamma_{\text{str}}]}$	0.811	0.811	0.811
Mean $\frac{\text{KL}_{\text{geo}}}{\text{KL}_{\text{str}}}$	0.900	0.900	0.900
Solver failures	1	1	1
Wilcoxon p -value	$< 10^{-14}$	$< 10^{-14}$	$< 10^{-14}$

exceeded 10^{-3} within the 200-iteration budget; all remaining 49 pairs are used in the reported means. This demonstrates practical reliability within the tested parameter range $\sigma \in (0.5, 2.5)$.

- **Path energy:** the geodesic achieves mean $\frac{E[\gamma_{\text{geo}}]}{E[\gamma_{\text{str}}]} = 0.811$ (18.9% energy reduction), consistent across all discretization levels $L \in \{64, 128, 256\}$.
- **Sequential KL cost:** mean $\frac{\text{KL}_{\text{geo}}}{\text{KL}_{\text{str}}} = 0.900$ (10.0% KL reduction), consistent across all L .

The geodesic achieves a mean sequential KL reduction of 10.0% (ratio $\frac{\text{KL}_{\text{geo}}}{\text{KL}_{\text{str}}} = 0.900$), consistent across $L \in \{64, 128, 256\}$. Because both the geodesic and the Euclidean straight-line path share the same endpoints, the endpoint divergence $D_{\text{KL}}(P_{\theta_1} \| P_{\theta_0})$ is identical; the reported 10.0% reduction therefore concerns the cumulative sequential surrogate (18), not the detectability of the final object under the standard endpoint-only threat model. Wilcoxon test: $p = 1.8 \times 10^{-15}$ for each L (Table 8, Figure 4).

8.7. Simulation S5: MNIST VAE Latent-Space Embedding

The previous simulations use low-dimensional parametric families where the Fisher metric, geodesics, and curvature are available in closed form or via standard ODE solvers. To test whether the geometric guidance remains meaningful for a learned generator, we train a small variational autoencoder [35] on MNIST with a 2-D latent space. The decoder $p_{\theta}(x | z)$ is Bernoulli over the 28×28 pixels, so the latent space inherits the decoder pullback Fisher metric

$$F(z) = \sum_{i=1}^{784} \frac{1}{\pi_i(z)(1 - \pi_i(z))} \nabla_z \pi_i(z) \nabla_z \pi_i(z)^{\top}. \quad (31)$$

Table 9

S5 – MNIST VAE embedding comparison (mean over three digit-class transitions). E is the discrete Fisher path energy; Acc. log-odds is the detector evidence accumulated over the trajectory.

Metric	Euclidean	NG path
Mean path energy E	3.07×10^3	1.17
Energy reduction	99.96%	
Mean accumulated log-odds	77.85	55.98
Detection reduction	25.0%	

A message is encoded by moving from a cover latent z_0 (the mean latent of one digit class) to a target latent z_1 (the mean latent of another class). We compare two embedding strategies:

- **Euclidean path:** linear interpolation $z(t) = (1-t)z_0 + tz_1$.
- **Natural-gradient (NG) path:** Fisher-preconditioned gradient descent on $\frac{1}{2}\|z - z_1\|^2$, which locally follows the decoder pullback metric and approximates the Fisher–Rao geodesic.

A small CNN steganalyzer is trained on Euclidean embeddings and then applied to both kinds of paths. Its output $p_{\text{stego}}(x)$ is converted to log-odds and accumulated along the trajectory, giving a sequential detection cost analogous to the sequential-KL surrogate used earlier.

The Euclidean baseline traverses latent regions where small parameter changes produce large changes in pixel probabilities; the Fisher pullback metric assigns high cost to those regions, so the Euclidean energy is orders of magnitude larger (Table 9). The NG path curves away from the high-sensitivity regions and yields a substantially lower accumulated detection score, even though the detector was trained exclusively on Euclidean embeddings (Figure 5). This confirms that the geometric guidance derived for simple exponential families remains useful when the generator is a learned deep network, and that the path itself — not only the endpoint — affects detectability.

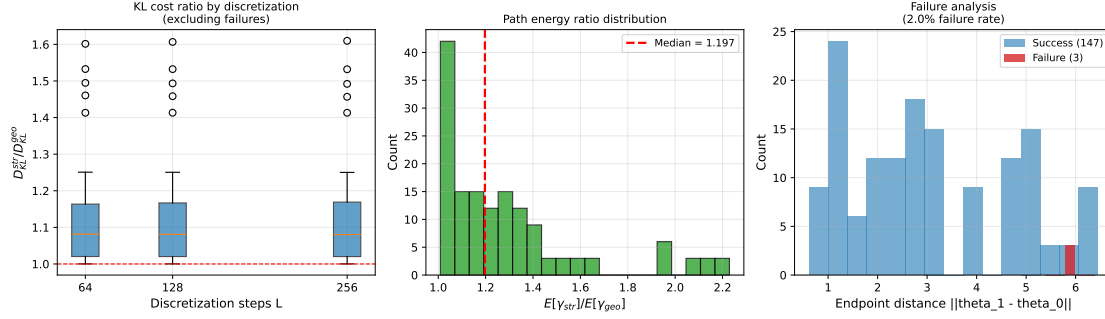


Figure 4: S4 – Path energy comparison for the 49 successful endpoint pairs out of 50 random trials. Each point represents one pair; values below the diagonal (dashed) favour the geodesic. The cluster near the origin corresponds to nearby endpoints where both paths perform similarly; the spread at larger distances confirms the 10.0% mean KL advantage of the geodesic (Table 8).

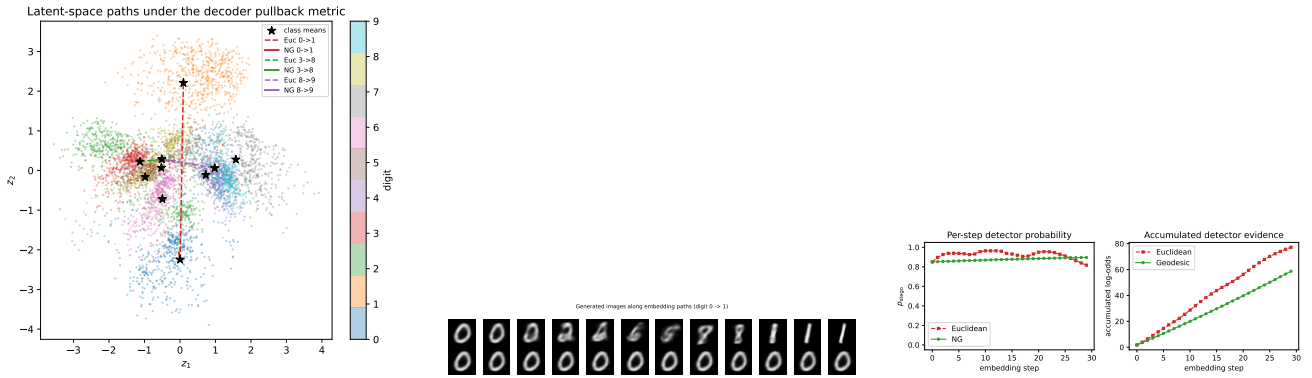


Figure 5: S5 – (Left) Latent-space paths under the decoder pullback metric for three digit-class transitions; the NG paths (solid) curve away from the high-Fisher regions traversed by the Euclidean baselines (dashed). (Middle) Generated images along the Euclidean (top) and NG (bottom) paths for the transition $0 \rightarrow 1$. (Right) Per-step detector probability and accumulated log-odds for the $0 \rightarrow 1$ transition; the NG path accumulates less detector evidence.

9. Detection-Theoretic Interpretation

9.1. Geometric Reformulation of Cachin’s Criterion

Theorem 7. Let $\gamma^* : [0, 1] \rightarrow S_G$ be the geodesic from θ_0 to θ_1 . In the local regime $d_g(\theta_0, \theta_1) \rightarrow 0$:

$$\varepsilon(\mathcal{E}_{\gamma^*}) = D_{\text{KL}}(P_{\theta_1} \| P_{\theta_0}) = \frac{1}{2} d_g(\theta_0, \theta_1)^2 + \frac{1}{6} T_{ijk} \Delta^i \Delta^j \Delta^k + O(d_g^4) \quad (32)$$

where $\Delta = \theta_1 - \theta_0$ and T_{ijk} is the Amari–Chentsov skewness tensor. For Gaussian families with mean-only parameterization, the cubic term vanishes and $D_{\text{KL}} = \frac{1}{2} d_g^2 + O(d_g^4)$.

Remark 12. Equation (32) is an asymptotic local identity derived from the Taylor expansion of the KL divergence. For exponential families where the cubic skewness term vanishes (e.g., Gaussian with mean-only parameterization), the leading term is $D_{\text{KL}} = \frac{1}{2} d_g^2 + O(d_g^4)$. The remainder is bounded by

$$|R_4| \leq \frac{1}{24} \max_{\theta \in [\theta_0, \theta_1]} |\partial_i \partial_j \partial_k \partial_l \psi(\theta)| \cdot d_g^4,$$

where ψ is the log-partition function.

From security level to geometric distance. For a given security level ε , the Fisher–Rao distance between cover and stego parameters follows from inverting the expansion:

$$d_g(\theta_0, \theta_1) = \sqrt{2\varepsilon} + O(\varepsilon^{3/2}).$$

For extremely small theoretical values such as $\varepsilon = 2^{-128}$, the formal expansion gives $d_g = 2^{-63.5} + O(2^{-192}) \approx 2^{-64}$, and the higher-order term becomes negligible under the stated smoothness and vanishing-skewness conditions. This higher-order simplification applies only when the cubic term vanishes. For families with non-zero skewness, such as GMMs, the cubic contribution must be retained and may dominate at intermediate distances.

9.2. The Steganalyzer as an m-Projection

We now introduce the adversary into the geometric framework. The purpose of this subsection is not to propose a new operational steganalysis algorithm, but to reinterpret the optimal Neyman–Pearson test in exponential families through the affine geometry of information geometry.

Throughout this subsection, we consider the binary testing problem

$$H_0 : X_1, \dots, X_n \sim P_{\theta_c}, \quad H_1 : X_1, \dots, X_n \sim P_{\theta_s},$$

where θ_c is the cover parameter and $\theta_s \neq \theta_c$ is the stego parameter. Type-I and type-II errors are defined as

$$\alpha_n = P_{\theta_c}(\text{reject } H_0), \quad \beta_n = P_{\theta_s}(\text{accept } H_0).$$

The following result gives an information-geometric form of the Neyman–Pearson test in exponential families. Its novelty is not that exponential families admit dual-affine coordinates (a classical result due to Amari and Nagaoka [5]), but that the *adversarial structure of the steganographic problem* endows this duality with a specific operational meaning: the embedder’s optimal transport path is governed by the Levi-Civita ($\nabla^{(0)}$, Fisher–Rao) geometry, while the optimal steganalyzer’s rejection region is an affine half-space in expectation coordinates, i.e. it is governed by the m -geometry ($\nabla^{(-1)}$). The normal to that half-space is the e -geodesic direction connecting θ_c and θ_s . Thus the ± 1 connection pair plays opposite and complementary roles for the two adversaries, a structural observation that we have not found in prior work on exponential families or on steganographic security.

Theorem 8. *Let $\{P_\theta : \theta \in \Theta\}$ be a regular exponential family with sufficient statistic T , log-partition function ψ , natural parameter θ , and expectation parameter $\eta(\theta) = \nabla\psi(\theta)$. Fix $\theta_c, \theta_s \in \Theta$ with $\theta_s \neq \theta_c$, and define*

$$\hat{\eta}_n = \frac{1}{n} \sum_{i=1}^n T(X_i).$$

Then the following statements hold.

- (i) Log-likelihood ratio decomposition. The normalized log-likelihood ratio satisfies

$$\Lambda_n := \frac{1}{n} \sum_{i=1}^n \log \frac{p_{\theta_s}(X_i)}{p_{\theta_c}(X_i)} = \langle \theta_s - \theta_c, \hat{\eta}_n - \eta_c \rangle - D_{\text{KL}}(P_{\theta_c} \| P_{\theta_s}), \quad (33)$$

where $\eta_c = \eta(\theta_c)$.

- (ii) Neyman–Pearson region as an m -flat half-space. For any threshold τ , the rejection region $\{\Lambda_n > \tau\}$ is equivalent to

$$\mathcal{R}_\tau = \{\hat{\eta}_n : \langle \theta_s - \theta_c, \hat{\eta}_n \rangle > c_\tau\}, \quad (34)$$

for a constant c_τ depending on τ , θ_c , and θ_s . Hence the Neyman–Pearson rejection boundary is affine in expectation coordinates, i.e., it is m -flat.

- (iii) Projection interpretation. In a dually flat exponential family, the e -geodesic joining θ_c to θ_s is the natural-parameter segment

$$\theta(t) = (1-t)\theta_c + t\theta_s, \quad t \in [0, 1].$$

The statistic $\langle \theta_s - \theta_c, \hat{\eta}_n \rangle$ orders empirical distributions according to their affine coordinate along the normal direction to the m -flat Neyman–Pearson boundary. Equivalently, the optimal test can be interpreted geometrically as comparing the empirical distribution with the e -geodesic direction connecting cover and stego parameters through the dual affine structure.

More explicitly, when the m -projection of the empirical distribution onto the e -geodesic segment exists and is unique, its coordinate

$$t^* = \arg \min_{t \in [0, 1]} D_{\text{KL}}(P_{\hat{\eta}_n} \| P_{\theta(t)})$$

is monotone with the Neyman–Pearson score in a neighbourhood of the segment. Thus thresholding the NP score is locally equivalent to thresholding this projection coordinate.

Proof. For an exponential family,

$$\log p_\theta(x) = \langle \theta, T(x) \rangle - \psi(\theta) + \log h(x).$$

Therefore,

$$\log \frac{p_{\theta_s}(x)}{p_{\theta_c}(x)} = \langle \theta_s - \theta_c, T(x) \rangle - (\psi(\theta_s) - \psi(\theta_c)).$$

Averaging over n samples gives

$$\Lambda_n = \langle \theta_s - \theta_c, \hat{\eta}_n \rangle - (\psi(\theta_s) - \psi(\theta_c)).$$

Using the exponential-family identity

$$D_{\text{KL}}(P_{\theta_c} \| P_{\theta_s}) = \psi(\theta_s) - \psi(\theta_c) - \langle \theta_s - \theta_c, \eta_c \rangle,$$

we obtain (33).

Part (ii) follows immediately because the condition $\Lambda_n > \tau$ is equivalent to a linear inequality in $\hat{\eta}_n$. Since expectation coordinates are m -affine coordinates, the rejection boundary is m -flat.

Part (iii) follows from the dual affine structure of exponential families: e -geodesics are affine in natural coordinates, whereas m -geodesics are affine in expectation coordinates. The Neyman–Pearson statistic is the linear functional in expectation coordinates associated with the natural direction $\theta_s - \theta_c$. The final statement is local because existence, uniqueness, and monotonicity of the m -projection coordinate can fail globally without additional convexity assumptions. $\square$

Steganographic consequence. The optimal detector is not an external add-on to the embedding model: in exponential-family signal models, its Neyman–Pearson statistic is aligned with the same cover-to-stego parameter displacement that defines the embedding problem. This gives a common statistical coordinate system for analyzing embedding design and steganalysis.

Remark 13 (What Theorem 9 adds beyond classical Stein’s lemma). *The classical Stein’s lemma identifies $-D_{\text{KL}}(P_{\theta_s} \| P_{\theta_c})$ as the optimal type-II error exponent for a simple binary hypothesis test. That result is well known. What is new here is the chain of connections it establishes in the steganographic setting: (i) the Stein exponent is expressed via the Fisher–Rao distance $d_g(\theta_c, \theta_s)$ through the local KL expansion (Theorem 7), so that the parameter-space geometry directly governs detection difficulty; and (ii) this connects to the embedder’s problem — a small Fisher–Rao distance buys both low accumulated embedding distortion (Theorem 3) and high type-II error for the steganalyzer. The local bound $\beta_n \lesssim \exp(-nd_g^2/2)$ thus closes the loop between the embedding geometry and the detection geometry, something that the endpoint-only security literature (Cachin, Fridrich–Ker) does not provide.*

Theorem 9. *Under the conditions of Theorem 8, the optimal type-II error at fixed type-I level $\alpha \in (0, 1)$ satisfies the Stein limit*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \beta_n^{(\alpha)} = -D_{\text{KL}}(P_{\theta_s} \| P_{\theta_c}). \quad (35)$$

Combined with Theorem 7, this gives

$$\beta_n^{(\alpha)} \lesssim \exp\left(-\frac{n}{2} d_g(\theta_c, \theta_s)^2 + O(n d_g(\theta_c, \theta_s)^3)\right), \quad (36)$$

so the type-II error decays exponentially in n at leading rate $\frac{1}{2} d_g(\theta_c, \theta_s)^2$. In the local regime, the squared Fisher–Rao distance therefore gives the leading-order per-observation detection cost of the embedding.

Proof. Equation (35) is the Chernoff–Stein lemma ([37], Thm. 12.8.1). Equation (36) follows by substituting the local approximation (32). $\square$ $\square$

Corollary 3. *For cover parameter θ_c , n observations, and required type-II error $\beta \in (0, 1)$, define the local asymptotic undetectability region*

$$\mathcal{U}_n(\theta_c, \beta) := \{\theta_s \in \Theta : \beta_n^{(\alpha)}(\theta_c, \theta_s) \geq \beta\}.$$

In the local regime $d_g(\theta_c, \theta_s) \rightarrow 0$, the Stein approximation implies the leading-order condition

$$d_g(\theta_c, \theta_s)^2 \lesssim \frac{2 \log(1/\beta)}{n}.$$

Thus, to leading order, the local Fisher–Rao ball

$$B_g\left(\theta_c, \sqrt{\frac{2 \log(1/\beta)}{n}}\right)$$

provides a conservative geometric proxy for the region in which the type-II error remains at least β .

The geometric quantity

$$C_{\text{geo}}^{(\text{Stein})} = \log_2 \text{vol}_g\left(B_g\left(\theta_c, \sqrt{\frac{2 \log(1/\beta)}{n}}\right)\right)$$

should be interpreted as a local Fisher-ball packing proxy. It does not, by itself, constitute an operational payload capacity unless a codebook, decoder, estimation model, and reliability criterion are specified.

9.3. Embedding Capacity and Geometry

Definition 19. *The local geometric packing proxy at security level ϵ is*

$$C_{\text{geo}} = \log_2 \text{vol}_g\left(B_g(\theta_0, \sqrt{2\epsilon})\right).$$

This quantity measures the logarithmic Fisher–Rao volume of a local distinguishability ball. It should not be interpreted as an operational steganographic payload capacity without an explicit codebook, decoder, sampling model, and reliability criterion.

Remark 14. *The positivity and magnitude of C_{geo} depend on the explicit geometry of (S_G, g) and must be evaluated numerically for non-flat manifolds. In particular, the sign of the scalar curvature $R(\theta)$ or the Ricci curvature does not determine, by monotonicity alone, whether $C_{\text{geo}} > 0$ or how large it is. For the Gaussian family the Poincaré half-plane isometry yields an explicit volume formula; for the Gaussian-mixture family we compute $\text{vol}_g(B_g)$ directly by numerical integration (Simulation S3, Section 8.5).*

For Riemannian manifolds, the Bishop–Gromov comparison theorem relates geodesic ball volume to Ricci curvature. We distinguish two cases:

- **Non-negative Ricci curvature** ($\text{Ric} \geq 0$): volumes are bounded above by the Euclidean comparison ($\text{vol}(B_g(p, r)) \leq V_0(r) = \omega_d r^d$).
- **Negative Ricci curvature** ($\text{Ric} \geq (d-1)K$ with $K < 0$): the comparison space is hyperbolic $\mathbb{H}^d(K)$, whose balls grow exponentially in r . The ratio $\text{vol}(B_g(p, r))/V_K(r)$ is non-increasing and bounded by 1, but $V_K(r) \gg V_0(r)$ for $K < 0$.

The GMM family of §4.3 has negative scalar curvature at interior parameter points ($\pi \approx 0.5$, $\Delta\mu \approx 1-3$), as confirmed by Simulation S3 ($R \in [-7.4, -2.3]$). Near degenerate configurations ($\pi \approx 0.1$ or 0.9 , $\Delta\mu \approx 0.3$, where $\kappa(F) \approx 96,528$), the scalar curvature turns strongly positive ($R \approx +325$), reflecting the spherical local geometry of ill-conditioned regimes. For the interior (negative- R) regime we apply the hyperbolic form:

Theorem 10. *Let $(\mathcal{M}, g)$ be a complete d -dimensional Riemannian manifold with $\text{Ric} \geq (d-1)K$ for some $K < 0$. Let $V_K(r)$ denote the volume of a geodesic ball of radius r in the d -dimensional hyperbolic space of constant sectional curvature K . Then for all $p \in \mathcal{M}$ and $r > 0$:*

$$\frac{\text{vol}(B_g(p, r))}{V_K(r)} \leq 1.$$

Consequently, the geodesic ball volume is bounded above by the hyperbolic comparison volume $V_K(r)$, which grows exponentially in r . This bound is not tight enough to infer increased capacity without additional upper bounds on Ricci curvature or explicit volume computation. For the low-dimensional families studied herein, ball volumes should be evaluated numerically (see Simulation S3).

Proof. The standard Bishop–Gromov theorem [38, Thm. III.4.2] states that if $\text{Ric} \geq (d-1)K$, then $\text{vol}(B_g(p, r))/V_K(r)$ is non-increasing in r , where $V_K(r)$ is the volume in the d -dimensional simply connected space form of constant sectional curvature K . For $K < 0$ this is hyperbolic space $\mathbb{H}^d(K)$, whose ball volume $V_K(r)$ grows exponentially in r . The ratio bound follows from monotonicity at $r \rightarrow 0$, where the ratio approaches 1. However, $\text{vol}(B_g(p, r)) \leq V_K(r)$ is an upper bound; it does not imply that the actual volume exceeds the Euclidean comparison $V_0(r)$. A lower bound on $\text{vol}(B_g(p, r))$ would require an upper bound on Ric , which is not provided by the Bishop–Gromov theorem. $\square$

Corollary 4. *The capacity formula*

$C_{\text{geo}} = \log_2 \text{vol}(B_g(\theta_0, \sqrt{2\epsilon}))$ *is valid whenever it yields a positive value. For a d -dimensional manifold with bounded sectional curvature $|K| \leq K_{\text{max}}$:*

$$C_{\text{geo}} > 0 \iff d \cdot \log_2(\sqrt{2\epsilon}) + \log_2 V_d > 0,$$

where V_d is the volume of the unit ball in $\mathbb{R}^d$. For $d \ll -2\log_2(\epsilon)$, this holds and the capacity is approximately

$$C_{\text{geo}} \approx \frac{d}{2} \log_2(2\epsilon) + \log_2 V_d + O(\epsilon \cdot K_{\text{max}}).$$

For $d \gg -2\log_2(\epsilon)$, the formula yields negative values and loses physical meaning; the regime is capacity-limited.

Remark 15. *The approximate formula above shows that, to leading order, C_{geo} depends on the dimension d and the security level ϵ , not on the sign of the curvature. The $O(\epsilon \cdot K_{\text{max}})$ correction is small in the local regime $\epsilon \ll 1$. Whether negative Ricci curvature increases or decreases capacity relative to the flat case must be determined by explicit numerical computation of $\text{vol}_g(B_g)$ for the specific manifold, not inferred from the Bishop–Gromov upper bound.*

Remark 16. *Combining Theorems 9, 10, and Corollary 2, we can now give a complete account of the role of curvature, which presents two opposite effects for the steganographer:*

1. **Capacity (detection perspective).** *The effect of negative Ricci curvature on embedding capacity cannot be inferred from lower Ricci bounds alone. Bishop–Gromov provides only an upper bound on ball volume via a hyperbolic comparison space; it does not guarantee that the actual volume exceeds the Euclidean case. For the GMM family, explicit numerical computation shows that Fisher–Rao ball volumes at interior points with negative scalar curvature remain within 0.3% of the Euclidean volume and strictly below the hyperbolic comparison bound, indicating that the hyperbolic upper bound is not attained. Capacity advantages, if any, must be verified by direct computation for the specific manifold.*
2. **Embedding stability (path perspective).** *Negative sectional curvature causes Jacobi fields to diverge (Theorem 6(iii)), making geodesic embedding paths sensitive to initial velocity perturbations. Negative curvature penalises the steganographer.*

These effects pull in opposite directions. For the GMM family, the interior regime ($\pi \approx 0.5$, $\Delta\mu \approx 1-3$, $R \in [-7, -2]$, $K_{\text{sec}}^{\text{min}} \approx -1.2$) keeps path instability bounded while not substantially altering capacity: the numerical volumes of Fisher–Rao balls remain within 0.3% of their Euclidean counterparts, so any capacity advantage of negative curvature is at most marginal. The extreme boundary ($\pi \approx 0.1$, $\Delta\mu \approx 0.3$, $R \approx +325$, $\kappa(F) \approx 96\,528$) has positive curvature, small undetectable balls, and an ill-conditioned Fisher metric, and is doubly unfavourable.

10. Discussion

10.1. Interpretation of Geodesic Embedding

The geodesic embedding path γ^* can be understood intuitively as the path that accumulates the least statistical information per unit length, creating the smallest cumulative sequential-KL footprint for a given endpoint separation. The 10.0% reduction of the sequential KL surrogate (S4) and the 2.1 $\times$ speedup (S2-EXT) provide numerical evidence that this geometric advantage can be observed in the tested moderate-dimensional families. As noted in Section 5.2, this does not imply a reduction of the endpoint divergence $D_{\text{KL}}(P_{\theta_1} \| P_{\theta_0})$ when only the final object is observed.

10.2. Path to Scalability for Deep Generators

The present framework is tractable for low-dimensional parametric families with available densities and scores ($d \leq 10$, e.g., Gaussian and GMM families). For deep generators with $d \sim 10^6-10^8$, three approximation avenues exist:

1. **Diagonal approximation.** Cost $O(d)$ per iteration, but error grows with the off-diagonal ratio ρ (Appendix F). For convolutional generators, ρ may be small due to spatial locality.
2. **K-FAC.** Kronecker-factored approximate curvature reduces cost to $O(d^{3/2})$ but assumes separable layer-wise structure, which may not hold for generators with skip connections or attention mechanisms.
3. **Online natural gradient.** Methods such as Ollivier [39] maintain a running estimate of F^{-1} with $O(d^2)$ memory, avoiding per-step inversion. The trade-off between approximation accuracy and forgetting factor remains to be analyzed for steganographic objectives.

Verdict. None of these approximations has been validated for steganographic generators. The 2.1 $\times$ speedup observed for the full Gaussian (S2-EXT) suggests that Fisher-aware updates may remain beneficial beyond the constant-metric case, but whether this advantage persists for deep architectures is an open empirical question.

10.3. Comparison with Adaptive Methods

The S2-ADAPTIVE simulation shows that, in the tested synthetic setting, NG-Stego achieves a lower mean iteration

count than Adam and RMSprop with their default hyper-parameters. A plausible explanation is that the natural gradient directly accounts for the *anisotropy* of the statistical manifold (captured by $F(\theta)$), while adaptive methods only estimate this anisotropy indirectly through gradient history.

10.4. Relation with Bregman Divergence

For exponential families, the KL divergence can be expressed as a *Bregman divergence* generated by the log-partition function ψ :

$$D_{\text{KL}}(P_\theta \| P_{\theta'}) = B_\psi(\theta' \| \theta) = \psi(\theta') - \psi(\theta) - \langle \theta' - \theta, \nabla \psi(\theta) \rangle. \quad (37)$$

Under this identification, the Fisher–Rao metric $g_{ij} = \partial_i \partial_j \psi$ is the Hessian of the Bregman generator, and the geodesics of the e -connection correspond to straight lines in the natural parameter θ (i.e., Bregman straight lines).

This duality has two immediate consequences for our framework. First, the one-step convergence of NG-Stego for diagonal exponential families (Theorem 5) can be understood as the Bregman proximal step being exact when ψ is quadratic. Second, the Detection Duality theorem (Theorem 8) admits a local Bregman-geometric interpretation: when the relevant projection exists, is unique, and is locally monotone with the Neyman–Pearson score, the test can be viewed as thresholding the projection coordinate of the empirical model onto the e -geodesic direction.

A full generalisation of Theorems 2 and 3 to arbitrary Bregman divergences (not necessarily arising from exponential families) is a natural extension; we leave it for future work.

10.5. Limitations and Scope

The present work has several important limitations.

First, most theoretical results are established under smooth exponential-family assumptions and may not directly extend to arbitrary deep generative models.

Second, the optimality results derived in Sections 5 and 7 rely on local second-order Fisher approximations of the KL divergence.

Third, the curvature-based analysis provides geometric indicators of embedding instability rather than direct operational local statistical surrogates.

Fourth, the current experiments remain limited to moderate-dimensional statistical manifolds.

Finally, the present paper focuses primarily on the signal-processing formulation and analysis of coverless steganographic embedding. A full operational evaluation against state-of-the-art image steganalyzers on large-scale datasets is left for future work.

10.6. Open Problems

1. Tight quantitative bound linking sectional curvature to KL divergence between distributions on neighboring geodesics (Remark 11).

2. Efficient estimation of the minimum sectional curvature $K_{\text{sec}}^{\min}(\theta)$ and scalar curvature $R(\theta)$ during generator training.
3. Global geodesic uniqueness for non-positively curved steganographic manifolds (Cartan–Hadamard analogue).
4. Game-theoretic interaction between geodesic encoding strategy and adaptive steganalysis.

11. Conclusion

We have developed a Fisher–Rao-guided signal-processing framework for generative steganographic embedding and detection, organized around five principal contributions.

(C1) A statistical signal-manifold formulation with proven geodesic completeness for exponential families (under the steepness condition on ψ), with uniqueness of minimizing geodesics established for the Gaussian family (via the Poincaré isometry giving $\kappa = -1/2$) and locally within the injectivity radius for general families. **(C2)** A local sequential-KL optimality result showing that Fisher–Rao geodesics minimize the leading-order accumulated KL distortion in the small-step regime, supported numerically by a 10.0% reduction of the sequential KL surrogate and an 18.9% energy reduction over the tested Euclidean coordinate paths. **(C3)** A natural-gradient formulation of steganographic embedding, extended beyond the diagonal case to general non-diagonal Fisher metrics, with convergence guarantees under geodesic strong-convexity and smoothness assumptions, and competitive empirical performance against Adam and RMSprop in the tested low-dimensional settings. **(C4)** A curvature-based instability analysis using two coordinate-invariant indices: the scalar curvature $R(\theta)$ and the minimum sectional curvature $K_{\text{sec}}^{\min}(\theta)$ (Monte-Carlo estimated over the Grassmannian), together with bootstrap confidence intervals. For the Gaussian family, $R = -1$ (constant); for GMMs, R varies across the parameter space. **(C5)** The *Detection Duality* theorem (Theorem 8): in regular exponential families, the Neyman–Pearson statistic admits a local dual-affine interpretation through the m -projection of the empirical distribution onto the e -geodesic direction connecting θ_c and θ_s . The Stein error exponent is $-D_{\text{KL}}(P_{\theta_s} \| P_{\theta_c})$, and in the local regime

$$D_{\text{KL}}(P_{\theta_s} \| P_{\theta_c}) = \frac{1}{2} d_g(\theta_c, \theta_s)^2 + O(d_g^3).$$

Consequently, a local Fisher ball of radius $\sqrt{2 \log(1/\beta)/n}$ provides a leading-order conservative proxy for parameter configurations whose type-II error remains above β .

Together, contributions (C1)–(C5) establish a local embedding-detection duality: the *embedder* is modeled through Fisher–Rao transport paths on the statistical manifold, while the *optimal Neyman–Pearson steganalyzer* admits a local dual-affine interpretation through projection onto the e -geodesic direction connecting cover and stego parameters. The structural ± 1 -duality under Amari–Chentsov connections appears in the affine description of the detection geometry,

whereas the embedding optimality result is governed by the Levi-Civita/Fisher–Rao geometry.

The revised security analysis (Section 9) corrects the Bishop–Gromov application for negative curvature regimes, provides explicit validity conditions for the capacity formula, and reveals the two-faces-of-curvature phenomenon: negative curvature yields a trade-off between *potential* Fisher-ball volume expansion (governed by the explicit geometry of the Fisher–Rao ball, not by comparison-theorem monotonicity) and geodesic transport instability (amplified Jacobi-field divergence, cost for the embedder).

Scope statement. The proposed methodology is currently validated on tractable statistical signal models, where full Fisher-geometric analysis is available for low-dimensional parametric families ($d \leq 10$). Extensions to deep generators with $d \sim 10^6$ – 10^8 require structured FIM approximations (diagonal, K-FAC, online NG) whose accuracy for steganographic objectives has not yet been validated empirically. The Detection Duality theorem applies to any exponential family; extending it to computationally bounded (neural) steganalyzers remains an open problem.

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A. Proof of Theorem 2 (Geodesic Completeness)

For an exponential family with natural parameter space Θ and log-partition function ψ , the Fisher metric is $g_{ij}(\theta) = \partial_i \partial_j \psi(\theta)$. Since ψ is strictly convex with super-linear growth at the boundary of Θ , Θ is a convex open subset of $\mathbb{R}^d$ and the metric g is the Hessian of a convex function.

For the Gaussian family $\mathcal{N}(\mu, \sigma^2)$, the change of coordinates $y = \sigma\sqrt{2}$ gives the metric $ds^2 = 2(d\mu^2 + dy^2)/y^2$, which is 2 times the Poincaré metric on $\mathbb{H}^2$ (see Appendix G). This metric is complete (do Carmo [25], Prop. 2.1), so geodesics exist between any two points. Non-positive curvature ($\kappa = -1/2$) and Cartan–Hadamard then imply uniqueness of geodesics for the Gaussian family.

For general exponential families, with open natural parameter space $\Theta \subseteq \mathbb{R}^d$, completeness follows from the super-linear growth of ψ at the boundary of Θ , which prevents geodesics from reaching the boundary in finite distance (Brown, 1986, Theorem 3.8). By Hopf–Rinow, metric completeness implies geodesic completeness. Uniqueness of minimizing geodesics holds locally within the injectivity radius; global uniqueness for arbitrary exponential families would require non-positive sectional curvature on all of Θ , which we do not assume here. $\square$

B. Note on Bishop–Gromov and Capacity

The Bishop–Gromov volume comparison theorem (Chavel [38], Theorem 10) states: if $\text{Ric} \geq (d-1)K$ on a complete d -dimensional Riemannian manifold $\mathcal{M}$, then for any $p \in \mathcal{M}$ and $0 < r < R$:

$$\frac{\text{vol}(B_g(p, R))}{\text{vol}(B_g(p, r))} \leq \frac{V_K(R)}{V_K(r)},$$

where $V_K(r)$ is the volume of a ball of radius r in the d -dimensional simply connected space form of constant sectional curvature K .

Taking $r \rightarrow 0$ gives $\text{vol}(B_g(p, R)) \leq V_K(R)$ for all $R > 0$. For $K < 0$, $V_K(R)$ grows exponentially (hyperbolic space), so the upper bound is very large. However, this does not guarantee that the actual volume $\text{vol}(B_g(p, R))$ is large; it only says it cannot exceed that large number. A lower

bound on volume would require an upper bound on Ricci curvature (i.e., $\text{Ric} \leq (d-1)K$), which Bishop–Gromov does not provide. Therefore, we cannot conclude that negative Ricci curvature *increases* capacity; we can only say that the theoretical maximum allowed volume is larger than in flat space. Any claim of increased capacity for a specific manifold requires explicit computation or a different comparison theorem. See Section 8.5 for explicit numerical confirmation on the GMM family.

In this paper, we do not rely on Bishop–Gromov to assert capacity advantages; we instead compute the actual Fisher ball volumes numerically for the low-dimensional families studied (Gaussian, GMM) and report them directly. For the GMM interior points where scalar curvature is negative, the computed ball volumes are modest and do not exhibit exponential growth. Hence the hyperbolic upper bound is not attained, and the capacity is determined by the actual geometry, not by the asymptotic bound.

C. Proof of Theorem 10

Proof. The standard Bishop–Gromov volume comparison theorem [38, Thm. III.4.2] states that if $\text{Ric} \geq (d-1)K$ on a complete d -dimensional Riemannian manifold $(\mathcal{M}, g)$, then for any $p \in \mathcal{M}$ and $0 < r < R$:

$$\frac{\text{vol}(B_g(p, R))}{\text{vol}(B_g(p, r))} \leq \frac{V_K(R)}{V_K(r)},$$

where $V_K(r)$ is the volume of a geodesic ball of radius r in the d -dimensional simply connected space form of constant sectional curvature K .

Taking $r \rightarrow 0$ and using $\text{vol}(B_g(p, r))/V_K(r) \rightarrow 1$ yields, for all $R > 0$:

$$\text{vol}(B_g(p, R)) \leq V_K(R). \quad (38)$$

For $K < 0$, the comparison space is hyperbolic space $\mathbb{H}^d(K)$, whose ball volume grows exponentially in R :

$$V_K(R) \sim c_d |K|^{-d/2} e^{(d-1)\sqrt{|K|}R} \quad \text{for large } R.$$

Thus $V_K(R)$ is very large, and the right-hand side of (38) exceeds the Euclidean comparison volume $V_0(R) = \omega_d R^d$.

However, (38) is an *upper* bound: it states that the actual volume $\text{vol}(B_g(p, R))$ cannot exceed a large number. It does not guarantee that the actual volume is itself large, nor that it exceeds the Euclidean volume $V_0(R)$. A lower bound on $\text{vol}(B_g(p, R))$ would require an *upper* bound on Ricci curvature (i.e. $\text{Ric} \leq (d-1)K$ with $K < 0$), which the Bishop–Gromov theorem does not provide.

Consequently, one cannot infer from $\text{Ric} \geq (d-1)K$ with $K < 0$ that the Fisher–Rao ball is larger than its Euclidean counterpart, nor that embedding capacity is increased. Any claim of capacity advantage for a specific statistical manifold requires explicit numerical computation of $\text{vol}_g(B_g)$ or a different comparison theorem yielding a lower bound. $\square$

D. Proof of Theorem 5 (NG Convergence Rate)

Let $\mathcal{L}(\theta) = D_{KL}(P_\theta \| P_{\theta^*})$ with θ^* the unique minimizer. Under assumptions (i)–(iii), the Riemannian gradient descent with step size $\eta = 1/L_g$ satisfies the standard contraction (Bonnabel [24], Thm. 3):

$$\mathcal{L}(\theta_{t+1}) \leq \mathcal{L}(\theta_t) \cdot \left(1 - \frac{\mu}{L_g}\right).$$

For exponential families with constant Fisher metric, the KL objective is quadratic in the corresponding affine coordinates and the natural-gradient step coincides with the exact Newton/Fisher-preconditioned step. In this special case, the Riemannian condition number is $\kappa_g = 1$, which yields one-step convergence for the mean-only Gaussian subfamily considered in Simulation S2.

For non-constant Fisher metrics, the Riemannian condition number $\kappa_g = L_g/\mu$ is generally larger than one and depends on the local geometry of the statistical manifold near θ^* . Euclidean gradient descent has an analogous contraction factor involving its Euclidean condition number $\kappa_e = L_e/\mu_e$. Therefore, the relative improvement of natural-gradient descent is controlled by the ratio κ_e/κ_g , not by $\kappa(F(\theta^*))$ alone. In particular, the speedup may be much smaller than $\kappa(F(\theta^*))$ when curvature or metric variation increases κ_g . This is consistent with Simulation S2-EXT, where the observed speedup is approximately $2.1\times$ despite $\kappa(F) \approx 15$. $\square$

E. Christoffel Symbols for the Gaussian Family

For the univariate Gaussian $\mathcal{N}(\mu, \sigma^2)$ with $\theta = (\mu, \sigma)$ and Fisher metric $g = \text{diag}(\sigma^{-2}, 2\sigma^{-2})$:

$$\begin{aligned} \Gamma_{11}^1 &= 0, & \Gamma_{12}^1 &= \Gamma_{21}^1 = -\sigma^{-1}, & \Gamma_{22}^1 &= 0, \\ \Gamma_{11}^2 &= \sigma^{-1}/2, & \Gamma_{12}^2 &= \Gamma_{21}^2 = 0, & \Gamma_{22}^2 &= -\sigma^{-1}. \end{aligned} \quad (39)$$

Geodesic equations: $\ddot{\mu} - 2\dot{\mu}\dot{\sigma}/\sigma = 0$, $\ddot{\sigma} + \dot{\mu}^2/(2\sigma) - \dot{\sigma}^2/\sigma = 0$.

F. Error Bound for Diagonal FIM Approximation

Let F be the exact FIM and $F^{\text{diag}} = \text{diag}(F_{11}, \dots, F_{dd})$. Let $\rho = \max_{i \neq j} |F_{ij}|/\sqrt{F_{ii}F_{jj}}$. The natural gradient is $\tilde{\nabla} = F^{-1}g$; the diagonal approximation is $\tilde{\nabla}^{\text{diag}} = (F^{\text{diag}})^{-1}g$. Then:

$$\frac{\|\tilde{\nabla}^{\text{diag}} - \tilde{\nabla}\|_g}{\|\tilde{\nabla}\|_g} \leq \frac{\rho d}{1 - \rho d} \quad \text{for } \rho < 1/d.$$

This follows from Neumann series expansion of F^{-1} with $F = F^{\text{diag}} + F^{\text{off}}$ where $\|(F^{\text{diag}})^{-1}F^{\text{off}}\|_2 \leq \rho d$.

G. Explicit Fisher Metric for the Univariate Gaussian

We derive the Fisher–Rao metric for $P_\theta = \mathcal{N}(\mu, \sigma^2)$ with $\theta = (\mu, \sigma)$, $\sigma > 0$, from first principles.

Score functions. The log-likelihood is

$$\ell(x, \theta) = -\frac{1}{2} \log(2\pi) - \log \sigma - \frac{(x - \mu)^2}{2\sigma^2}.$$

The score vector is

$$\partial_\mu \ell = \frac{x - \mu}{\sigma^2}, \quad \partial_\sigma \ell = -\frac{1}{\sigma} + \frac{(x - \mu)^2}{\sigma^3}.$$

Fisher information matrix. Using $\mathbb{E}[x - \mu] = 0$, $\mathbb{E}[(x - \mu)^2] = \sigma^2$, and $\mathbb{E}[(x - \mu)^4] = 3\sigma^4$:

$$\begin{aligned} g_{11} &= \mathbb{E}[(\partial_\mu \ell)^2] = \mathbb{E}\left[\left(\frac{x - \mu}{\sigma^2}\right)^2\right] = \frac{1}{\sigma^2}, \\ g_{22} &= \mathbb{E}[(\partial_\sigma \ell)^2] = \mathbb{E}\left[\left(-\frac{1}{\sigma} + \frac{(x - \mu)^2}{\sigma^3}\right)^2\right] \\ &= \mathbb{E}\left[\frac{1}{\sigma^2} - \frac{2(x - \mu)^2}{\sigma^4} + \frac{(x - \mu)^4}{\sigma^6}\right] \\ &= \frac{1}{\sigma^2} - \frac{2\sigma^2}{\sigma^4} + \frac{3\sigma^4}{\sigma^6} = \frac{2}{\sigma^2}, \\ g_{12} &= g_{21} = \mathbb{E}[\partial_\mu \ell \partial_\sigma \ell] \\ &= \mathbb{E}\left[\frac{x - \mu}{\sigma^2} \left(-\frac{1}{\sigma} + \frac{(x - \mu)^2}{\sigma^3}\right)\right] \\ &= -\frac{1}{\sigma^3} \mathbb{E}[x - \mu] + \frac{1}{\sigma^5} \mathbb{E}[(x - \mu)^3] = 0. \end{aligned}$$

Hence the Fisher–Rao metric tensor is

$$g = \frac{1}{\sigma^2} d\mu \otimes d\mu + \frac{2}{\sigma^2} d\sigma \otimes d\sigma. \quad (40)$$

Isometry to the Poincaré half-plane. Under the change of variable $y = \sigma\sqrt{2}$ (so $\sigma = y/\sqrt{2}$):

$$g = \frac{2(d\mu^2 + dy^2)}{y^2},$$

which is 2 times the standard Poincaré metric on $\mathbb{H}^2 = \{(\mu, y) : y > 0\}$. Under a constant rescaling $g \rightarrow \lambda g$, sectional curvature transforms as $K \rightarrow K/\lambda$. The standard Poincaré metric has $K = -1$; with scale factor $\lambda = 2$, the sectional curvature of our metric is $\kappa = -1/2$ everywhere (constant, independent of σ). The scalar curvature in dimension $d = 2$ is $R = 2K = 2 \cdot (-1/2) = -1$, a constant independent of the parameters (μ, σ) .

Christoffel symbols. From (40) with $g_{11} = \sigma^{-2}$, $g_{22} = 2\sigma^{-2}$, $g_{12} = 0$, one computes:

$$\begin{aligned} \Gamma_{12}^1 &= \Gamma_{21}^1 = -\sigma^{-1}, & \Gamma_{11}^1 &= \Gamma_{22}^1 = 0, \\ \Gamma_{11}^2 &= \frac{1}{2}\sigma^{-1}, & \Gamma_{22}^2 &= -\sigma^{-1}, & \Gamma_{12}^2 &= \Gamma_{21}^2 = 0. \end{aligned}$$

(41)

These agree with Appendix E. The geodesic equations are $\ddot{\mu} - 2\dot{\mu}\dot{\sigma}/\sigma = 0$ and $\ddot{\sigma} + \dot{\mu}^2/(2\sigma) - \dot{\sigma}^2/\sigma = 0$, whose solutions are vertical lines ($\mu = \text{const}$) and semicircles centred on the $\sigma = 0$ axis (do Carmo [25], Boothby [40]).